

TEST REPORT

For

Zhejiang Qibin Technology Co.,Ltd.

INVERTER

Model(s):

HYBRID PUMP INVERTER AC&DC:

7.5KW LX100-7R5G-4-PV

2.2KW LX100-2R2G-SS2-PV

3KW LX100-003G-SS2-PV

4KW LX100-004G-SS2-PV

5.5KW LX100-5R5G-SS2-PV

7.5KW LX100-7R5G-SS2-PV

11KW LX100-011G-SS2-PV

5.5KW LX100-5R5G-4-PV

7.5KW LX100-7R5G-4-PV

11KW LX100-011G-4-PV

15KW LX100-015G-4-PV

18KW LX100-018G-4-PV

22KW LX100-022G-4-PV

30KW LX100-030G-4-PV

37KW LX100-037G-4-PV

45KW LX100-045G-4-PV

55KW LX100-055G-4-PV

75KW LX100-075G-4-PV

90KW LX100-090G-4-PV

110KW LX100-110G-4-PV

132KW LX100-132G-4-PV

160KW LX100-160G-4-PV

185KW LX100-185G-4-PV

200KW LX100-200G-4-PV

SOLAR PUMP INVERTER:

7.5KW LX100-7R5G-4-PV

2.2KW LX100-2R2G-SS2-PV

3KW LX100-003G-SS2-PV

4KW LX100-004G-SS2-PV

5.5KW LX100-5R5G-SS2-PV

7.5KW	LX100-7R5G-SS2-PV
11KW	LX100-011G-SS2-PV
5.5KW	LX100-5R5G-4-PV
7.5KW	LX100-7R5G-4-PV
11KW	LX100-011G-4-PV
15KW	LX100-015G-4-PV
18KW	LX100-018G-4-PV
22KW	LX100-022G-4-PV
30KW	LX100-030G-4-PV
37KW	LX100-037G-4-PV
45KW	LX100-045G-4-PV
55KW	LX100-055G-4-PV
75KW	LX100-075G-4-PV
90KW	LX100-090G-4-PV
110KW	LX100-110G-4-PV
132KW	LX100-132G-4-PV
160KW	LX100-160G-4-PV
185KW	LX100-185G-4-PV
200KW	LX100-200G-4-PV

Prepared For: Zhejiang Qibin Technology Co.,Ltd.
No.528 Hedian Avenue, Qinshan Street, Haiyan County,
Jiaxing City, Zhejiang Province

Prepared By: Zhejiang NewRay Test&Certification Technology Co.,Ltd.
5th Floor,No.9 Fengling Road,Wuchang Street,Yuhang
District, Hangzhou City,Zhejiang



Report No.: ZNCT250108004
Date of Test: Jan.07-08,2025
Date of report: Jan.08,2025

TEST REPORT

Report No.:	ZNCT250108004
Date of issue.....:	Jan.08,2025
Total number of pages.....:	65
Testing Laboratory:	Zhejiang NewRay Test&Certification Technology Co.,Ltd.
Address.....:	5th Floor,No.9 Fengling Road,Wuchang Street,Yuhang District, Hangzhou City,Zhejiang
Applicant name:	Zhejiang Qibin Technology Co.,Ltd.
Address.....:	No.528 Hedian Avenue, Qinshan Street, Haiyan County, Jiaxing City, Zhejiang Province
Testing specification:	
Standards.....:	IEC 62109-1:2010 IEC 62109-2:2011
Non-standard test method.....:	N/A
Test Report Form No:	--
Test Report Form(s) Originator.....:	
Master TRF.....:	Dated: 2022-07
This report is specially limited to the above client company and product model only. It may not be duplicated without prior written consent of ZNCT.	
Test item description:	INVERTER
Trade Mark.....:	N/A
Manufacturer.....:	Zhejiang Qibin Technology Co.,Ltd. No.528 Hedian Avenue, Qinshan Street, Haiyan County, Jiaxing City, Zhejiang Province
Test Model.....:	7.5KW LX100-7R5G-4-PV
Specifications.....:	POWER:7.5KW INPUT: 420~720VDC OUTPUT: 3PH AC 0-380V ,18.5A 0-600HZ

Test procedure and location:**Testing Laboratory.....: Zhejiang NewRay Test&Certification Technology Co.,Ltd.****Address.....: 5th Floor, No.9 Fengling Road, Wuchang Street, Yuhang District, Hangzhou City, Zhejiang****Date of Issue.....: Jan.08,2025****Tested by (name + signature)..... : Shelly***Shelly***Reviewed by (name + signature)..... : Eric.Zhang***Eric. Zhang***Approved by (name + signature).....: Jeny***Jeny*

List of Attachments (including a total number of pages in each attachment):
Summary of Test:

Test performed (name of test and test clause):

- IEC 62109-1:2010
- IEC 62109-2:2011

Testing location:

5th Floor, No.9 Fengling Road, Wuchang Street, Yuhang District, Hangzhou City, Zhejiang

The submitted samples were found to comply with the requirements of above specification.

No EUROPEAN GROUP DIFFERENCES AND NATIONAL DIFFERENCES

Copy of marking plate:
INVERTER

Model(s): 7.5KW LX100-7R5G-4-PV



Manufacturer: Zhejiang Qibin Technology Co., Ltd.

Address: No.528 Hedian Avenue, Qinshan Street, Haiyan County, Jiaxing City, Zhejiang Province

Remark on above marking:

- 1, The height of CE symbols is more than 5 mm;
- 2, The height of WEEE symbols is more than 7 mm;

Test item particulars..... :	
Classification of installation and use.....:	
Supply Connection.....:	
Possible assessment case verdicts:	
- assessment case does not apply to the object.....: N (Not applicable) - assessment object does meet the requirement.....: P (Pass) - assessment object does not meet the requirement...: F (Fail)	
Date of receipt of test item..... :	Jan.06,2025
Date (s) of performance of test..... :	Jan.07-08,2025
General remarks:	
"(See Enclosure #)" refers to additional information appended to the report. "(See appended table)" refers to a table appended to the report. Throughout this report a comma / point is used as the decimal separator. Clause numbers between brackets refer to clauses in 62109	
Manufacturer's Declaration per sub-clause 4.2.5 of IEC 60335-1:	
The application for obtaining a Certificate includes more than one factory location and a declaration from the Manufacturer stating that the sample(s) submitted for evaluation is (are) representative of the products from each factory has been provided.....:	Not applicable
General product information:	

Part I : IEC 62109-1 report

4	GENERAL TESTING REQUIREMENTS		P
4.1	General		P
4.2	General conditions for testing		P
4.2.1	Sequence of tests		P
4.2.2	Reference test conditions		P
4.2.2.1	Environmental conditions		P
4.2.2.2	State of equipment		P
4.2.2.3	Position of equipment	Be fixed in accordance with the manufacturer's instruction	P
4.2.2.4	Accessories	No accessories or operator interchangeable parts	N/A
4.2.2.5	Covers and removable parts		N/A
4.2.2.6	Mains supply a) Voltage: b) Frequency: c) Polarity: Earthing: Over-current Protection:		P
4.2.2.7	Supply ports other than the mains		P
4.2.2.7.1	Photovoltaic supply sources a) Open circuit voltage: b) Short-circuit current:	Test with a source capable of 1.25 times the PCE rated maximum input current.	P
4.2.2.7.2	Battery inputs		N/A
4.2.2.8	Conditions of loading for output ports		P
4.2.2.9	Earthing terminals		P
4.2.2.10	Controls	No such controls	N/A
4.2.2.11	Available short circuit current		P
4.3	Thermal testing		P
4.3.1	General		P
4.3.2	Maximum temperatures	See appended table	P
4.3.2.1	General		P
4.3.2.2	Touch temperatures		P
4.3.2.3	Temperature limits for mounting surfaces		P
4.4	Testing in single fault condition		P
4.4.1	General		P
4.4.2	Test conditions and duration for testing under fault conditions		P
4.4.2.1	General		P
4.4.2.2	Duration of tests		P
4.4.3	Pass/fail criteria for testing under fault conditions		P
4.4.3.1	Protection against shock hazard		P

4.4.3.2	Protection against the spread of fire		P
4.4.3.3	Protection against other hazards		P
4.4.3.4	Protection against parts expulsion hazards		P
4.4.4	Single fault conditions to be applied		P
4.4.4.1	Component fault tests	(see appended table)	P
4.4.4.2	Equipment or parts for short-term or intermittent operation	Not for short-term or intermittent operation	N/A
4.4.4.3	Motors	No motors	N/A
4.4.4.4	Transformer short circuit tests		P
4.4.4.5	Output short circuit		P
4.4.4.6	Backfeed current test for equipment with more than one source of supply		P
4.4.4.7	Output overload		P
4.4.4.8	Cooling system failure		P
4.4.4.9	Heating devices	No heating devices	N/A
4.4.4.10	Safety interlock systems		N/A
4.4.4.11	Reverse d.c. connections	.	P
4.4.4.12	Voltage selector mismatch	No voltage selector	N/A
4.4.4.13	Mis-wiring with incorrect phase sequence or polarity		P
4.4.4.14	Printed wiring board short-circuit test	No insulation distance less than the required spacing.	N/A
4.5	Humidity preconditioning	(see appended table 7.5)	P
4.5.1	General		P
4.5.2	Conditions	92.5% R.H. 40°C.	P
4.6	Backfeed voltage protection		P
4.6.1	Backfeed tests under normal conditions	1.25V	P
4.6.2	Backfeed tests under single-fault conditions	1.25V	P
4.6.3	Compliance with backfeed tests		P
4.7	Electrical ratings tests	(see appended table 4.7)	P
4.7.1	Input ratings		P
4.7.1.1	Measurement requirements for DC input ports		P
4.7.2	Output ratings		P
5	MARKING AND DOCUMENTATION		P
5.1	Marking		--
5.1.1	General		P
	Equipment shall bear markings as specified in 5.1 and 5.2	Label are marked on PCE and graphic symbol is explained in user manual	P
	Graphic symbols may be used and shall be in accordance with Annex C or IEC 60417 as applicable.		P

	Graphic symbols shall be explained in the documentation provided with the PCE.		P
5.1.2	Durability of markings		P
	Markings required by this clause to be located on the PCE shall remain clear and legible under conditions of NORMAL USE and resist the effects of cleaning agents specified by the manufacturer		P
5.1.3	Identification		P
	The equipment shall, as a minimum, be permanently marked with:	The marking remains clear and legible after test.	P
	a) the name or trade mark of the manufacturer or supplier	Zhejiang Qibin Technology Co.,Ltd.	P
	b) model number, name or other means to identify the equipment	7.5KW LX100-7R5G-4-PV Single-phase	P
	c) a serial number, code or other marking allowing identification of manufacturing location and the manufacturing batch or date within a three- month time period.	Within three months	P
5.1.4	Equipment ratings	See below	P
	Unless otherwise specified in another part of IEC 62109, the following ratings, as applicable shall be	Special requirement as IEC/IEC 62109-2	P
	marked on the equipment:		
	- input voltage, type of voltage (a.c. or d.c.),frequency, and max. continuous current for each input	Refer to the marking label	P
	–output voltage, type of voltage (a.c. or d.c.), frequency, max. continuous current, and for a.c. outputs, either the power or power factor for each output	Refer to the marking label	P
	–the ingress protection (IP) rating as in 6.3 below	IP 30	P
5.1.5	Fuse identification		N/A
	Marking shall be located adjacent to each fuse or fuseholder, or on the fuseholder, or in another location provided that it is obvious to which fuse the marking applies, giving the fuse current rating and where fuses of different voltage rating value could be fitted, the fuse voltage rating.		P
	Where fuses with special fusing characteristics such as time delay or breaking capacity are necessary, the type shall also be indicated		N/A

	For fuses not located in operator access areas and for soldered-in fuses located in operator access areas, it is permitted to provide an unambiguous cross-reference (for example, F1, F2, etc.) to the servicing instructions which shall contain the relevant information.		P
5.1.6	Terminals, Connections, and Controls		P
	If necessary for safety, an indication shall be given of the purpose of Terminals, connectors, controls, and indicators, and their various positions, including any connections for coolant fluids such as water and drainage. The symbols in Annex C may be	“+” and “-” marked close to DC input connect. “L” “N” and “GND” marked close to AC output terminal	P
	used, and where there is insufficient space, symbol 9 of Annex C may be used.	block.	
	Push-buttons and actuators of emergency stop devices, and indicator lamps used only to indicate a warning of danger or the need for urgent action shall be coloured red.	No such device.	N/A
	A multiple-voltage unit shall be marked to indicate the particular voltage for which it is set when shipped from the factory. The marking is allowed to be in the form of a paper tag or any other non- permanent material.		P
	A unit with d.c. terminals shall be plainly marked indicating the polarity of the connections, with:	See below	P
	- the sign “+” for positive and “-”, for negative; or		P
	- a pictorial representation illustrating the proper polarity where the correct polarity can be unambiguously determined from the representation	Not provided	N/A
5.1.6.1	Protective Conductor Terminals		P
	The means of connection for the protective earthing conductor shall be marked with:	The protective earthing terminal is connected.	P
	- symbol 7 of Annex C; or		P
	- the letters “PE”; or		N/A
	- the colour coding green-yellow.		N/A
5.1.7	Switches and circuit-breakers		N/A
	The on and off-positions of switches and circuits breakers shall be clearly marked. If a push-button switch is used as the power switch, symbols 10 and 16 of Annex C may be used to indicate the on- position, or symbols 11 and 17 to indicate the off- position, with the pair of symbols (10 and 16, or 11 and 17) close together.		N/A
5.1.8	Class II Equipment	Class II	P

	Equipment using Class II protective means throughout shall be marked with symbol 12 of Annex C. Equipment which is only partially protected by DOUBLE INSULATION or REINFORCED INSULATION shall not bear symbol 12 of Table Annex C.		P
	Where such equipment has provision for the connection of an earthing conductor for functional reasons (see 7.3.6.4) it shall be marked with symbol 6 of Annex C		N/A
5.1.9	Terminal boxes for External Connections	No such terminal box	N/A
	Where required by note 1 of Table 2 as a result of high temperatures of terminals or parts in the wiring compartment, there shall be a marking, visible beside the terminal before connection, of either:		N/A
	a) the minimum temperature Rating and size of the cable to be connected to the TERMINALS; or		N/A
	b) a marking to warn the installer to consult the installation instruction. Symbol 9 of Table D-1 is an acceptable marking		N/A
5.2	Warning markings		P
5.2.1	Visibility and legibility requirements for warning markings		P
	Warning markings shall be legible, and shall have minimum dimensions as follows:		P
	- Printed symbols shall be at least 2,75 mm high		P
	- Printed text characters shall be at least 1.5 mmhigh and shall contrast in colour with the background		P
	- Symbols or text that are moulded, stamped orengraved in a material shall have a character height of at least 2,0 mm, and if not contrasting in colour from the background, shall have a depth or raised height of at least 0,5 mm.		P
	If it is necessary to refer to the instruction manual to preserve the protection afforded by the equipment, the equipment shall be marked with symbol 9 of Annex C	The manual provides necessary information for warning marking	P
	Symbol 9 of Annex C is not required to be used adjacent to symbols that are explained in the manual		P
5.2.2	Content for warning markings		P
5.2.2.1	Ungrounded heat sinks and similar parts	Grounded heatsink and metal enclosure	N/A

	An ungrounded heat sink or other part that may be mistaken for a grounded part and involves a risk of electric shock in accordance with 7.3 shall be marked with symbol 13 of Annex C, or equivalent. The marking may be on or adjacent to the heat sink and shall be clearly visible when the PCE is disassembled to the extent that a risk of contact with the heat sink exists.		N/A
5.2.2.2	Hot Surfaces		P
	A part of the PCE that exceeds the temperature limits specified in 4.3.2 shall be marked with symbol 14 of Annex C or equivalent.		P
5.2.2.3	Coolant	Coolant is not used	N/A
	A unit containing coolant that exceeds 70 °C shall be legibly marked externally where readily visible after installation with symbol 15 of Annex C. The documentation shall provide a warning regarding the risk of burns from hot coolant, and either:		N/A
	a) statement that coolant system servicing is to be done only by SERVICE PERSONNEL, or		N/A
	b) instructions for safe venting, draining, or otherwise working on the cooling system, if these operations can be performed without OPERATOR		N/A
	access to HAZARDS internal to the equipment		
5.2.2.4	Stored energy		P
	Where required by 7.3.9.2 or 7.4.2 the PCE shall be marked with Symbol 21 of Annex C and the time to discharge capacitors to safe voltage and energy levels shall accompany the symbol.		P
5.2.2.5	Motor guarding		N/A
	Where required by 8.2 a marking shall be provided where it is visible to service personnel before removal of a guard, warning of the hazard and giving instructions for safe servicing (for example disconnection of the source before removing the guard).	No motor inside enclosure	N/A
5.2.3	Sonic hazard markings and instructions	Hazardous noise level not produced	N/A
	If required by 10.2.1 a PCE shall:		N/A
	a) be marked to warn the operator of the sonic pressure hazard; or		N/A

	b) be provided with installation instructions that specify how the installer can ensure that the sound pressure level from equipment at its point of use after installation, will not reach a value, which could cause a hazard. These instructions shall include the measured sound pressure level, and shall identify readily available and practicable protective materials or measures which may be used.		N/A
5.2.4	Equipment with multiple sources of supply		P
	A PCE with connections for multiple energy sources shall be marked with symbol 13 of Annex C and the manual shall contain the information required in 5.3.4.		P
	The symbol shall be located on the outside of the unit or shall be prominently visible behind any cover giving access to hazardous parts.		P
5.2.5	Excessive touch current		N/A
	Where required by 7.3.6.3.7 the PCE shall be marked with symbol 15 of Annex C. See also 5.3.2 for information to be provided in the installation manual.	The touch current does not exceed limited	N/A
5.3	Documentation		P
5.3.1	General		P
	The documentation provided with the PCE shall provide the information needed for the safe operation, installation, and (where applicable) maintenance of the equipment. The documentation shall include the items required in 5.3.2 through 5.3.4, and the following:		P
	a) explanations of equipment makings, including symbols used		P
	b) location and function of terminals and controls		P
	c) all ratings or specifications that are necessary to safely install and operate the PCE, including the following environmental ratings along with an explanation of their meaning and any resulting installation requirements:		P
	– ENVIRONMENTAL CATEGORY as per 6.1	Indoor use only.	P
	– WET LOCATIONS classification for the intended external environment as per 6.1	Not suitable for wet location.	N/A
	– POLLUTION DEGREE classification for the intended external environment as per 6.2		N/A
	– INGRESS PROTECTION rating as per 6.3	IP 30	P
	– Ambient temperature and relative humidity ratings	-20℃~+50℃, < 95%RH	P
	– MAXIMUM altitude rating	Up to 5000 m	P

	– OVERVOLTAGE CATEGORY assigned to each input and output port as per 7.3.7.1.2, accompanied by guidance regarding how to ensure that the installation complies with the required overvoltage categories;	OVC II	P
	d) a warning that when the photovoltaic array is exposed to light, it supplies a d.c. voltage to the PCE		P
5.3.1.1	Language	English provide	P
	Instructions related to safety shall be in a language that is acceptable in the country where the equipment is to be installed.	For other country language further evaluated is needed	N/A
5.3.1.2	Format		P
	In general, the documentation must be provided in printed form and is to be delivered with the equipment.	Printed form provided	P
	For equipment which requires the use of a computer for both installation and operation, documentation may be provided in electronic format without accompanying printed format.		N/A
5.3.2	Information related to installation		P
	The documentation shall include installation and where applicable, specific commissioning instructions and, if necessary for safety, warnings against hazards which could arise during installation or commissioning of the equipment. The information provided shall include:		P
	a) assembly, location, and mounting requirements:		P
	b) ratings and means of connection to each source of supply and any requirements related to wiring and external controls, colour coding of leads, disconnection means, or overcurrent protection needed, including instructions that the installation position shall not prevent access to the		P
	disconnection means;		
	c) ratings and means of connection of any outputs from the PCE, and any requirements related to wiring and external controls, colour coding of leads, or overcurrent protection needed;		P
	d) explanation of the pin-out of connectors for external connections, unless the connector is used for a standard purpose (e.g. RS 232)		P
	e) ventilation requirements;		P
	f) requirements for special services, for example cooling liquid;	No special services	N/A

	g) instructions and information relating to sound pressure level if required by 10.2.1;		N/A
	h) where required by 14.8.1.3, instructions for the adequate ventilation of the room or location in which PCE containing vented or valve- regulated batteries is located, to prevent the accumulation of hazardous gases;	No such battery	N/A
	i) tightening torque to be applied to wiring terminals;		N/A
	j) values of backfeed short-circuit currents available from the PCE on input and output conductors under fault conditions, if those currents exceed the max. rated current of the circuit, as per 4.4.4.6;	Not exceeds the max. rated current.	N/A
	k) for each input to the PCE, the max value of short-circuit current available from the source, for which the PCE is designed; and		P
	l) compatibility with RCD and RCM;		N/A
	m) instructions for protective earthing, including the information required by 7.3.6.3.7 if a second protective earthing conductor is to be installed:	Touch current not exceeds limit.	N/A
	n) where required by 7.3.8, the installation instructions shall include the following or equivalent wording:		N/A
	“This product can cause a d.c. current in the external protective earthing conductor. Where a residual current-operated protective (RCD) or monitoring (RCM) device is used for protection in a case of direct or indirect contact, only an RCD or RCM of Type B is allowed on the supply side of this product.”		N/A
	o) for PCE intended to charge batteries, the battery nominal voltage rating, size, and type		N/A
	p) PV array configuration information, such as ratings, whether the array is to be grounded or floating, any external protection devices needed, etc.		N/A
5.3.3	Information related to operation		P
	Instructions for use shall include any operating		P
	instructions necessary to ensure safe operation, including the following, as applicable:		
	- Instructions for adjustment of controls including the effects of adjustment;		P
	- Instructions for interconnection to accessories and other equipment, including indication of suitable accessories, detachable parts and any special materials;		P

	- Warnings regarding the risk of burns from surfaces permitted to exceed the temperature limits of 4.3.2 and required operator actions to reduce the risk; and		N/A
	- Instructions, that if the equipment is used in a manner not specified by the manufacturer, the protection provided by the equipment may be impaired.		P
5.3.4	Information related to maintenance		P
	Maintenance instructions shall include the following:		P
	- Intervals and instructions for any preventivemaintenance that is required to maintain safety (for example air filter replacement or periodic re-tightening of terminals);		P
	- Instructions for accessing operator accessareas, if any are present, including a warning not to enter other areas of the equipment;		N/A
	- Part numbers and instructions for obtaining anyrequired operator replaceable parts;	No such part	N/A
	- Instructions for safe cleaning (if recommended)		N/A
	- Where there is more than one source of supplyenergizing the PCE, information shall be provided in the manual to indicate which disconnect device or devices are required to be operated in order to completely isolate the equipment.		P
5.3.4.1	Battery maintenance	No battery inside	N/A
	Where required by 14.8.5, the documentation shall include the applicable items from the following list of instructions regarding maintenance of batteries:		N/A
	- Servicing of batteries should be performed orsupervised by personnel knowledgeable about batteries and the required precautions		N/A
	- When replacing batteries, replace with thesame type and number of batteries or battery packs		N/A
	- General instructions regarding removal andinstallation of batteries		N/A
	- CAUTION: Do not dispose of batteries in a fire.The batteries may explode.		N/A
	- CAUTION: Do not open or damage batteries.Released electrolyte is harmful to the skin and eyes. It may be toxic.		N/A
	- CAUTION: A battery can present a risk		N/A
	ofelectrical shock and high short-circuit current. The following precautions should be observed when working on batteries:		

	a) Remove watches, rings, or other metal objects.		N/A
	b) Use tools with insulated handles.		N/A
	c) Wear rubber gloves and boots.		N/A
	d) Do not lay tools or metal parts on top of batteries		N/A
	e) Disconnect charging source prior to connecting or disconnecting battery terminals		N/A
	f) Determine if battery is inadvertently grounded. If inadvertently grounded, remove source from ground. Contact with any part of a grounded battery can result in electrical shock. The likelihood of such shock can be reduced if such grounds are removed during installation and maintenance (applicable to equipment and remote battery supplies not having a grounded supply circuit).		N/A
6	ENVIRONMENTAL REQUIREMENTS AND CONDITIONS		P
	The manufacturer shall rate the PCE for the following environmental conditions:		P
	– ENVIRONMENTAL CATEGORY, as in 6.1 below	Indoor use	P
	– Suitability for WET LOCATIONS or not		N/A
	– POLLUTION DEGREE rating in 6.2 below		N/A
	– INGRESS PROTECTION (IP) rating, as in 6.3 below	IP 20	P
	– Ultraviolet (UV) exposure rating, as in 6.4 below	No	N/A
	– Ambient temperature and relative humidity ratings, as in 6.5 below	-20℃-50℃, < 95%RH	P
6.1	Environmental categories and minimum environmental conditions		P
6.1.1	Outdoor	No	N/A
6.1.2	Indoor, unconditioned		P
6.1.3	Indoor, conditioned		N/A
6.2	Pollution degree		N/A
6.3	Ingress Protection	IP 20	P
6.4	UV exposure	No	N/A
6.5	Temperature and humidity	-20℃- 50℃, < 95%RH	P
7	PROTECTION AGAINST ELECTRIC SHOCK AND ENERGY HAZARDS		P
7.1	General		P
7.2	Fault conditions	Normal and single fault condition are considered	P
7.3	Protection against electric shock		P
7.3.1	General		P
7.3.2	Decisive voltage classification		P
7.3.2.1	Use of decisive voltage class (DVC)	Working voltage and protective measure and considered.	P
7.3.2.2	Limits of DVC (according table 6)		P

7.3.2.3	Short-terms limits of accessible voltages under fault conditions		P
7.3.2.4	Requirements for protection (according table 7)	The touchable interface are separated by reinforced insulation provided.	P
7.3.2.5	Connection to PELV and SELV circuits		N/A
7.3.2.6	Working voltage and DVC		P
7.3.2.6.1	General	Transients and voltage fluctuation are disregarded. And worst case normal operation condition is considered.	P
7.3.2.6.2	AC working voltage (see Figure 2)		P
7.3.2.6.3	DC working voltage (see Figure 3)		P
7.3.2.6.4	Pulsating working voltage (see Figure 4)		N/A
7.3.3	protective separation	In the PCE the earthed metal enclosure is evaluated by means of basic insulation from DVC C circuit The touchable interface are separated by reinforced insulation provided by the isolating transformer.	P
	Protective separation shall be achieved by:		P
	§ double or reinforced insulation, or	By isolating transformer.	P
	§ protective screening, i.e. by a conductivescreen connected to earth by protective bonding in the PCE, or connected to the protective earth conductor itself, whereby the screen is separated from live parts by at least basic insulation, or		N/A
	§ protective impedance comprising limitation ofcurrent per 7.3.5.3 and of discharged energy per 7.3.5.4, or		N/A
	§ limitation of voltage according to 7.3.5.4.		N/A
	The protective separation shall be fully and effectively maintained under all conditions of intended use of the PCE		P
7.3.4	Protection against direct contact	Enclosure provided	P
7.3.4.1	General		P
	Protection against direct contact is employed to prevent persons from touching live parts that do not meet the requirements of 7.3.5 and shall be provided by one or more of the measure given in 7.3.4.2 (enclosures and barriers) and 7.3.4.3 (insulation).	Enclosure provided	P

	Open type sub-assemblies and devices do not		N/A
	require protective measures against direct contact but the instruction provided with the equipment must indicate that such measures must be provided in the end equipment or in the installation.		
	Product intended for installation in CLOSED ELECTRICAL OPERATING AREAS, (see 3.9) need not have protective measures against direct contact, except as required by 7.3.4.2.4.		N/A
7.3.4.2	Protection by means of enclosures and barriers		P
	The following requirements apply where protection against contact with live parts is provided by enclosures or barriers, not by insulation in accordance with 7.3.4.3.	Enclosure provided to prevent access to inside live parts	P
7.3.4.2.1	General		P
	Parts of enclosures and barriers that provide protection in accordance with these requirements shall not be removable without the use of a tool (see 7.3.4.2.3).	Secured by screws	P
	Polymeric materials used to meet these requirements shall also meet the requirements of 13.6		P
7.3.4.2.2	Access probe criteria		P
	Protection is considered to be achieved when the separation between the test probes and live parts, when tested as described below, is as follows:		P
	a) decisive voltage classification A, (DVC A) - the probe may touch the live parts	The interface is considered as DVC A	P
	b) decisive voltage classification B, (DVC B) - the probe must not touch bare live parts	The DVC B circuit is not accessible by probe	P
	c) decisive voltage classification C, (DVC C) – the probe must have adequate clearance to live parts, based on the clearance for Basic insulation using the recurring peak working voltage involved,	The DVC C circuit is not accessible by probe	N/A
7.3.4.2.3	Access probe tests		P
	Compliance with 7.3.4.2.1 is checked by all of the following:		P
	a) Inspection; and		P

	Tests with the test finger (Figure D.1) and test pin (Figure D.2) of 0E, the results of which shall comply with the requirements of 7.3.4.2.1 a), b), and as applicable. Probe tests are performed on openings in the enclosures after removal of parts that can be detached or opened by an operator without the use of a tool, including fuseholders, and with operator access doors and covers open. It is permitted to leave lamps in place for this test. Connectors that can be separated by an operator without use of a tool, shall also be tested during and after disconnection. Any movable parts are to be put in the most unfavourable position.		P
	The test finger and the test pin are applied as above, without appreciable force, in every possible position, except that floor-standing equipment having a mass exceeding 40 kg is not tilted.		P
	Equipment intended for building-in or rack mounting, or for incorporation in larger equipment, is tested with access to the equipment limited according to the method of mounting detailed in the installation instructions.		N/A
	c) Openings preventing the entry of the jointed test finger (Figure E-1 of 0E) during test b) above, are further tested by means of straight unjointed test finger (Figure E-3 of 0E), applied with a force of 30 N. If the unjointed finger enters, the test with the jointed finger is repeated except that the finger is applied using any necessary force up to 30 N.		N/A
	d) In addition to a) – c) above, top surfaces of enclosure shall be tested with the IP3X probe of IEC 60529. The test probe shall not penetrate the top surface of the enclosure when probed from the vertical direction $\pm 5^\circ$ only.		N/A
7.3.4.2.4	Service access areas	Inside PCE are not intentionally touched with energized part when installation and maintenance. Symbol 21 of Annex C are marked on PCE and explained in user manual	P
7.3.4.3	Protection by means of insulation of live parts	The earthed enclosure is with basic insulation form the live parts inside	P
	Where the requirements of 7.3.4.2 are not met, live parts shall be provided with insulation if:		P
	– their working voltage is greater than the maximum limit of decisive voltage class A, or		P

	– for a DVC A or B circuit, protective separation from adjacent circuit of DVC C is not provided (see note “±” under Table 7)		P
7.3.5	Protection in case of direct contact	The interfaces are direct contact and evaluated with reinforced insulation from live part.	P
7.3.5.1	General		P
	Protection in case of direct contact is required to ensure that contact with live parts does not produce a shock hazard.		P
	The protection against direct contact according to 7.3.4 is not required if the circuit contacted is separated from other circuits according to 7.3.2.3, and:	Considered	P
	– is of decisive voltage class A and complies with	The interfaces are DVC A and reinforced insulation from the	P
	7.3.5.2, or	live part by means of isolation transformer and optocoupler	
	– is provided with protective impedance according to 7.3.5.3, or		N/A
	– is limited in voltage according to 7.3.5.4		N/A
	In addition to the measures as given in 7.3.5.2 to 7.3.5.4, it shall be ensured that in the event of error or polarity reversal of connectors no voltages that exceed DVC A can be connected into a circuit with protective separation. This applies for example to plug-in-sub-assemblies or other plug-in devices which can be plugged-in without the use of a tool (key) or which are accessible without the use of a tool.	Considered	P
	Conformity is checked by visual inspection and trial insertion.		P
7.3.5.2	Protection using decisive voltage class A	The touchable interfaces are considered as DVC A and reinforced insulation from the live part by means of isolation transformer and optocoupler	P
7.3.5.3	Protection by means of protective impedance	Protective impedance not used as protective separation in the PCE	N/A
	Circuits and conductive parts do not require protection against direct contact if any connection to circuits of DVC-B or DVC-C is through protective impedance, and the accessible circuit or part is otherwise provided with protective separation from circuits of DVC-B or DVC-C		N/A

	according 7.3.3.		
7.3.5.3.1	Limitation of current through protective impedance		N/A
	The current available through protective impedance to earth and between simultaneously accessible parts, measured at the accessible live parts, shall not exceed a value of 3,5 mA a.c. or 10 mA d.c. under normal and single-fault conditions.		N/A
7.3.5.3.2	Limitation of discharging energy through protective impedance		N/A
	The discharging energy available between simultaneously accessible parts protected by protective impedance shall not exceed the charging voltage and capacitance limits given in Table 9, which applies to both wet and dry locations, under normal and single fault conditions. Refer to figure 8.		N/A
7.3.5.4	Protection by means of limited voltages		N/A
	That portion of a circuit that has its voltage reduced to DVC-A by a voltage divider that complies with the following requirements, and that is otherwise provided with protective separation from circuits of DVC-B or DVC-C according to 7.3.3, does not require protection against direct contact.		N/A
	The voltage divider shall be designed so that under normal and single fault conditions, including faults in the voltage division circuit, the voltage across the output of the voltage divider does not exceed the limit for DVC-A.		N/A
	This type of protection shall not be used in case of protective class II or unearthed circuits, because it relies on protective earth being connected.		N/A
7.3.6	Protection against indirect contact		P
7.3.6.1	General		P
	Protection against indirect contact is required to prevent shock- hazardous current being accessible from conductive parts during an insulation failure. This protection shall comply with the requirements for protective class I (basic insulation plus protective earthing), class II (double or reinforced insulation) or class III (limitation of voltages)	Class II	P
	That part of a PCE meets the requirements of 7.3.6.2 and 7.3.6.3 is defined as protective class I	The earthed metal enclosure meets this requirement	P

	That part of a PCE meets the requirements of 7.3.6.4 is defined as protective class II.		P
	That part of PCE which meets the requirements of decisive voltage class A and in which no hazardous voltages are derived, is defined as protective class III. No shock hazard is present in such circuits.		N/A
	Where protection against indirect contact is dependent on means provided during installation, the installation instructions shall provide details of the required means and shall indicate the associated hazards.		N/A
7.3.6.2	Insulation between live parts and accessible conductive parts		P
	Accessible conductive parts of equipment shall be separated from live parts by insulation meeting the requirements of Table 7 or by clearances as specified in 7.3.7.4 and creepages as specified in 7.3.7.5	See Cl. 7.3.7.4 and Cl. 7.3.7.5	P
7.3.6.3	Protective class I – Protective bonding and earthing		N/A
7.3.6.3.1	General		N/A
	Equipment of protective class I shall be provided with protective earthing, and with protective bonding to ensure electrical contact between accessible conductive parts and the means of connection for the external protective earthing conductor, except bonding is not required for:		N/A
	a) accessible conductive parts that are protected by one of the measures in 7.3.5.2 to 7.3.5.4, or		N/A
	b) accessible conductive parts are separated from live parts of DVC-B or -C using double or reinforced insulation.		N/A
7.3.6.3.2	Requirements for protective bonding		N/A
	Electrical contact with the means of connection of the external protective earthing conductor shall be achieved by one or more of the following means:		N/A
	a) through direct metallic contact;		N/A
	b) through other conductive parts which are not removed when the PCE or sub-units are used as intended;		N/A
	c) through a dedicated protective bonding conductor;		N/A
	d) through other metallic components of the PCE		N/A
	Where direct metallic contact is used and one or both of the parts involved is painted or coated, the paint or coating shall be removed in the area of contact, or reliably penetrated, to ensure metal to metal contact.	The metal enclosure is reliably penetrated and earthed	N/A

	For moving or removable parts, hinges or sliding contacts designed and maintained to have a low resistance are examples of acceptable means if they comply with the requirements of 7.3.6.3.3.	No such design	N/A
	Metal ducts of flexible or rigid construction and metallic sheaths shall not be used as protective bonding conductors, unless the device or material has been investigated as suitable for protective bonding purposes.	No such design	N/A
7.3.6.3.3	Rating of protective bonding		N/A
	Protective bonding shall withstand the highest thermal and dynamic stresses that can occur to the PCE item(s) concerned when they are subjected to a fault connecting live parts to accessible conductive parts. The protective bonding shall remain effective for as long as a fault to the accessible conductive parts persists or until an upstream protective device removes power from the part.		N/A
	Protective bonding shall meet following requirements:		N/A
	a) For PCE with an overcurrent protective device rating of 16 A or less, the impedance of the protective bonding means shall not exceed 0,1 Ω during or at the end of the test below.		N/A
	b) For PCE with an overcurrent protective device rating of more than 16 A, the voltage drop in the protective bonding test shall not exceed 2,5 V during or at the end of the test below.		N/A
	As alternative to a) and b) the protective bonding may designed according to the requirements for the external protective earthing conductor in 7.3.6.3.5, in which case no testing is required.		N/A
	The impedance of protective bonding means shall		N/A
	be checked by passing a test current through the bond for a period of time as specified below. The test current is based on the rating of the overcurrent protection for the equipment or part of the equipment under consideration, as follows:		
	a) For pluggable equipment type A, the overcurrent protective device is that provided external to the equipment (for example, in the building wiring, in the mains plug or in an equipment rack);		N/A

	b) For pluggable equipment type B and fixed equipment, the maximum rating of the overcurrent protective device specified in the equipment installation instructions to be provided external to the equipment;		N/A
	c) For a circuit or part of the equipment for which an overcurrent protective device is provided as part of the equipment, the rating of the provided overcurrent device.		N/A
	Voltages are measured from the protective earthing terminal to all parts whose protective bonding means are being considered. The impedance of the protective earthing conductor is not included in the measurement. However, if the protective earthing conductor is supplied with the equipment, it is permitted to include the conductor in the test circuit but the measurement of the voltage drop is made only from the main protective earthing terminal to the accessible part required to be earthed.	Measured form the farthest part of earthed metal enclosure to the input earth terminal	N/A
	On equipment where the protective earth connection to a subassembly or to a separate unit is part of a cable that also supplies power to that subassembly or unit, the resistance of the protective bonding conductor in that cable is not included in the protective bond impedance measurements for the subassembly or separate unit, as shown in Figure 11. However, this option is only permitted if the cable is protected by a suitably rated protective device that takes into account the size of the conductor. Otherwise the impedance of the protective bonding conductor between the separate units is to be included, by measuring to the protective earthing terminal where the power source enters the first unit in the system, as shown in Figure 12.	No separate unit provided	N/A
7.3.6.3.3.1	Test current, duration, and acceptance criteria		N/A
	The test current, duration of the test and acceptance criteria are as follows:		N/A
	a) For PCE with an overcurrent protective device rating of 16 A or less, the test current is 200% of the overcurrent protective device rating, but not less than 32 A, applied for 120s. The impedance of the protective bonding means during and at the end of the test shall not exceed 0,1 Ω .		N/A

	b) For PCE with an overcurrent protective device rating of more than 16 A, the test current is 200% of the overcurrent protective device rating and the duration of the test is as shown in Table 10 below. The voltage drop in the protective bonding means, during and at the end of the test, shall not exceed 2,5 V.	0.11V Test condition 25A, 2min	N/A
	c) During and after the test, there shall be no melting, loosening, or other damage that would impair the effectiveness of the protective bonding means.		N/A
	The test current is derived from an a.c or d.c supply source, the output of which is not earthed.		N/A
	As an alternative to Table 10, where the time- current characteristic of the overcurrent protective device that limits the fault current in the protective bonding means is known because the device is either provided in the equipment or fully specified in the installation instructions, the test duration may be based on that specific device's time-current characteristic,. The tests are conducted for a duration corresponding to the 200% current value on the time-current characteristic.		N/A
7.3.6.3.4	Protective bonding impedance (routine test)		N/A
	If the continuity of the protective bonding is achieved at any point by a single means only (for example a single conductor or single fastener), or if the PCE is assembled at the installation location, then the impedance of the protective bonding shall also be tested as a routine test. The test shall be as in 7.3.6.3.3, except for the following:		N/A
	§ the test current may be reduced to any convenient value greater than 10 A sufficient to allow measurement or calculation of the impedance of the protective bonding means:		N/A
	§ the test duration may be reduced to no less than 2 s		N/A
	For equipment subject to the type test in 7.3.6.3.3.1a), the impedance during the routine test shall not exceed 0,1Ω.		N/A
	For equipment subject to the type test in 7.3.6.3.3.1b) the impedance during the routine test shall not exceed 2,5 V divided by the test current required by 7.3.6.3.3.1b).		N/A
7.3.6.3.5	External protective earthing conductor		N/A

	A protective earthing conductor shall be connected at all times when power is supplied to PCE of protective class I. Unless local wiring regulations state otherwise, the protective earthing conductor cross-sectional area shall be determined from Table 11 or by calculation according to IEC 60364- 5-54.	Table 11 used. 10AWG and not less than phase conductor	N/A
	If the external protective earthing conductor is routed through a plug and socket or similar means of disconnection, it shall not be possible to disconnect it unless power is simultaneously removed from the part to be protected.	Permanently connected	N/A
	The cross-sectional area of every external protective earthing conductor which does not form part of the supply cable or cable enclosure shall, in any case, be not less than:		N/A
	§ 2,5 mm ² if mechanical protection is provided;		N/A
	§ 4 mm ² if mechanical protection is not provided.	The installation manual indicates the grounding wire have to stay consistent with Recommended wire for AC output.	N/A
	For cord-connected equipment, provisions shall be made so that the external protective earthing conductor in the cord shall, in the case of failure of the strain-relief mechanism, be the last conductor to be interrupted.		N/A
7.3.6.3.6	Means of connection for the external protective earthing conductor		N/A
7.3.6.3.6.1	General		N/A
	The means of connection for the external protective earthing conductor shall be located near the terminals for the respective live conductors. The means of connections shall be corrosion-resistant and shall be suitable for the connection of cables according to 7.3.6.3.5. The means of connection for the protective earthing conductor shall not be used as a part of the mechanical assembly of the equipment or for other connections. A separate means of connection shall be provided for each external protective earthing conductor. Connection and bonding points shall be so designed that their current-carrying capacity is not impaired by mechanical, chemical, or electrochemical influences. Where enclosures and/or conductors of aluminium or aluminium alloys are used, particular attention should be given to the problems of electrolytic corrosion.		N/A

	The means of connection for the protective earthing conductor shall be permanently marked with:		N/A
	symbol 7 of Annex C; or		N/A
	the colour coding green-yellow		N/A
	Marking shall not be done on easily changeable parts such as screws.		N/A
7.3.6.3.7	Touch current in case of failure of the protective earthing conductor		N/A
	The requirements of this sub-clause shall be		N/A
	satisfied to maintain safety in case of damage to or disconnection of the protective earthing conductor.		
	For pluggable equipment type A, the touch current measured in accordance with 7.5.4 shall not exceed 3,5 mA a.c. or mA d.c.		N/A
	For all other PCE, one or more of the following measure shall be applied, unless the touch current measured in accordance with 7.5.4 using the test network of IEC 60990 test figure 4 shall not exceed 3,5 mA a.c. or 10 mA d.c.	1.92 mA a.c. max	N/A
	a) Permanently connected wiring, and:	Not exceed 3.5mA a.c.	N/A
	a cross-section of the protective earthing conductor of at least 10 mm ² Cu or 16 mm ² Al; or		N/A
	automatic disconnection of the supply in case of discontinuity of the protective earthing conductor; or		N/A
	provision of an additional terminal for a second protective earthing conductor of the same cross- sectional area as the original protective earthing conductor and installation instruction requiring a second protective earthing conductor to be installed or		N/A
	b) Connection with an industrial connector according to IEC 60309 and a minimum protective earthing conductor cross-section of 2,5 mm ² as part of a multi-conductor power cable. Adequate strain relief shall be provided.	Not exceed 3.5mA a.c.	N/A
	In addition, the caution symbol 15 of Annex C shall be fixed to the product and the installation manual shall provide details of the protective earthing measures required in the installation as required in 5.3.2.		N/A
	When it is intended and allowed to connect two or more PCEs in parallel using one common PE conductor, the above touch current requirements apply to the maximum number of the PCEs to be connected in parallel, unless one of the measures in a)		N/A

	or b) above is used. The maximum number of parallel PCEs is used in the testing and has to be stated in the installation manual.		N/A
7.3.6.4	Protective Class II – Double or Reinforced Insulation	The touchable interfaces are evaluated with reinforced insulation form live parts inside	P
	Equipment or parts of equipment designed for protective class II shall have insulation between live parts and accessible surfaces in accordance with 7.3.4.3. The following requirements also apply:		P
	equipment designed to protective class II shallnot have means of connection for the external protective earthing conductor. However this does not apply if the external protective earthing		P
	conductor is passed through the equipment to equipment series-connected beyond it. In the latter event, the external protective earthing conductor and its means for connection shall be insulated with basic insulation from the accessible surface of the equipment and from circuits that employ protective separation, extra-low voltage, protective impedance and limited discharging energy, according to 7.3.5. This basic insulation shall correspond to the rated voltage of the series-connected equipment;		
	metal-encased equipment of protective class II may have provision on its enclosure for the connection of an equipotential bonding conductor;		N/A
	equipment of protective class II may haveprovision for the connection of an earthing conductor for functional reasons or for damping of overvoltages; it shall, however, be insulated as though it is a live part;		N/A
	equipment employing protective class II shallbe marked according to 5.1.8.		N/A
7.3.7	Insulation Including Clearance and Creepage Distance		P
7.3.7.1	General		P
	This subclause gives minimum requirements for insulation, based on the principles of IEC 60664.		P
	Manufacturing tolerances shall be taken into account during measurement of creepage, clearance, and insulation distance in the PCE.		P
	Insulation shall be selected after consideration of the following influences:		P
	pollution degree	PD2	P
	overvoltage category	OVC II	P

	supply earthing system		N/A
	insulation voltage	input: max. 60Vdc	P
	location of insulation	See table 7.3.7.	P
	type of insulation	See table 7.3.7	P
	Compliance of insulation, creepage distances, and clearance distances, shall be verified by measurement or visual inspection, and the tests of 7.5.		P
7.3.7.1.3	Supply earthing systems		N/A
	Three basic types of earthing system are described in IEC 60364-1. They are:		N/A
	TN system: has one point directly earthed, the accessible conductive parts of the installation being connected to that point by protective conductors. Three types of TN systems, TN-C, TN-S and TN-C-S, are defined according to the arrangement of the neutral and protective conductor.		N/A
	TT system: has one point directly earthed, the accessible conductive parts of the installation being connected to earth electrodes electrically independent of the earth electrodes of the power system;		N/A
	IT system: has all live parts isolated from earth or one point connected to earth through an impedance, the accessible conductive parts of the installation being earthed independently or collectively to the earthing system.		N/A
7.3.7.1.4	Insulation voltages	See table 7.3.7.	P
	Table 12 makes use of the circuit system voltage and overvoltage category to define the impulse withstands voltage and the temporary overvoltage.		P
7.3.7.2	Insulation between a circuit and its surroundings		P
7.3.7.2.1	General		P
7.3.7.2.2	Circuits connected directly to the mains		P
7.3.7.2.3	Circuits other than mains circuits		P
7.3.7.2.4	Insulation between circuits		P
7.3.7.3	Functional insulating		P
7.3.7.4	Clearance distances	(see appended table 7.3.7)	P
7.3.7.4.1	Determination		P
7.3.7.4.2	Electric field homogeneity		N/A
7.3.7.4.3	Clearance to conductive enclosures		P
7.3.7.5	Creepage distances	(see appended table 7.3.7)	P
7.3.7.5.1	General		P

7.3.7.5.2	Voltage		P
7.3.7.5.3	Materials		P
7.3.7.6	Coating	No coating provided insulation	N/A
7.3.7.7	PWB spacings for functional insulating	Functional insulation on PWB is comply with 7.3.7.4 and 7.3.7.5	P
7.3.7.8	Solid insulating		P
7.3.7.8.1	General		P
7.3.7.8.2	Requirements for electrical withstand capability of solid insulation		P
7.3.7.8.2.1	Basic, supplemental, reinforced, and double insulation	Refer to 7.5.2	P
7.3.7.8.2.2	Functional insulation		P
7.3.7.8.3	Thin sheet or tape material		P
7.3.7.8.3.1	General		P
7.3.7.8.3.2	Material thickness not less than 0,2 mm	Impulse test and voltage test are considered for insulation as basic insulation	P
7.3.7.8.3.3	Material thickness less than 0,2 mm		N/A
7.3.7.8.3.4	Compliance		N/A
7.3.7.8.4	Printed wiring boards		P
7.3.7.8.4.1	General		N/A
7.3.7.8.4.2	Use of coating materials		N/A
7.3.7.8.5	Wound components	Varnish are not considered as insulation	P
7.3.7.8.6	Potting materials		N/A
7.3.7.9	Insulation requirements above 30 kHz		N/A
7.3.8	Residual Current-operated protective (RCD) or monitoring (RCM) device compatibility		N/A
	RCD and RCM are used to provide protection against insulation faults in some domestic and industrial installations, additional to that provided by the installed equipment.		N/A
7.3.9	Capacitor discharge		P
7.3.9.1	Operator access area	Accessible interfaces are DVA circuit	P
	Equipment shall be so designed that there is no risk of electric shock in operator access areas from charge stored on capacitors after disconnection of the PCE.		P
7.3.9.2	Service access areas	Inside capacitor discharge to DVC A and no energy hazard level within 55mins	P

	Capacitors located behind panels that are removable for servicing, installation, or disconnection shall present no risk of electric shock or energy hazard from charge stored on capacitors after disconnection of the PCE.	Warning symbol 21 of Annex C is marked with 55mins.	P
7.4	Protection against energy hazards		P
7.4.1	Determination of hazardous energy level		P
	A hazardous energy level is considered to exist if		N/A
	a) The voltage is 2 V or more, and power available after 60 s exceeds 240 VA.		N/A
	b) The stored energy in a capacitor is at a voltage. U of 2 V or more, and the stored energy. E, calculated from the following equation, exceeds 20J: $E = 0,5 CU^2$		N/A
7.4.2	Operator Access Areas	No energized parts accessible by user	P
	Equipment shall be so designed that there is no risk of energy hazard in operator access areas from accessible circuits.		P
7.4.3	Services Access Areas		P
7.5	Electrical tests related to shock hazard	(see appended table 7.5)	P
7.5.1	Impulse voltage test (type test)	(see appended table 7.5)	P
7.5.2	Voltage test (dielectric strength test)	(see appended table 7.5)	P
7.5.2.1	Purpose of test		P
7.5.2.2	Value and type of test voltage		P
7.5.2.3	Humidity pre-conditioning		P
7.5.2.4	Performing the voltage test		P
7.5.2.5	Duration of the a.c. or d.c. voltage test		P
7.5.2.6	Verification of the a.c. or d.c. voltage test		P
7.5.3	Partial discharge test	The voltage stress on the insulation is not greater than 700V.	N/A
7.5.4	Touch current measurement (type test)		P
	The touch current shall be measured if required by 7.3.6.3.7 and shall not be greater than 3.5 mA a.c. or 10 mA d.c. or special measures of protection as given in 7.3.6.3.7 are required.		P
	For type tests on PCE for which wet locations requirements apply according to 6.1, the humidity pre-conditioning of 4.5 shall be performed immediately prior to the touch current test.		P
7.5.5	Equipment with multiple sources of supply		P
8	PROTECTION AGAINST MECHANICAL HAZARDS		P

8.1	General		--
	Operation shall not lead to a mechanical HAZARD in NORMAL CONDITION or SINGLE FAULT CONDITION. Edges, projections, corners, openings, guards, handles and the like, that are accessible to the operator shall be smooth and rounded so as not to cause injury during normal use of the equipment.		P
	Conformity is checked as specified in 8.2 to 8.6.		P
8.2	Moving parts		N/A
	Moving parts shall not be able to crush, cut or pierce parts of the body of an OPERATOR likely to contact them, nor severely pinch the OPERATOR's skin. Hazardous moving parts of equipment, that is moving parts which have the potential to cause injury, shall be so arranged, enclosed or guarded as to provide adequate protection against the risk of personal injury.	No moving parts	N/A
8.2.1	Protection of service persons		P
	Protection shall be provided such that unintentional contact with hazardous moving parts is unlikely during servicing operations. If a guard over a hazardous moving part may need to be removed for servicing, the marking of symbol 15 of Table D-1 shall be applied on or near the guard.		P
8.3	Stability		N/A
	Equipment and assemblies of equipment not secured to the building structure before operation shall be physically stable in NORMAL USE.	Wall mounted	N/A
8.4	Provisions for lifting and carrying		P
	If carrying handles or grips are fitted to, or supplied with, the equipment, they shall be capable of withstanding a force of four times the weight of the equipment.		P
	Equipment or parts having a mass of 18 kg or more shall be provided with a means for lifting and carrying or directions shall be given in the manufacturer's documentation.	Not exceed 18kg	N/A
8.5	Wall mounting		N/A
	Mounting brackets on equipment intended to be mounted on a wall or ceiling shall withstand a force of four times the weight of the equipment.		N/A
8.6	Expelled parts		N/A
	Equipment shall contain or limit the energy of parts that could cause a HAZARD if expelled in the event of a fault.		N/A

9	PROTECTION AGAINST FIRE HAZARDS		P
9.1	Resistance to fire		P
	This subclause specifies requirements intended to reduce the risk of ignition and the spread of flame, both within the equipment and to the outside, by the appropriate use of materials and components and by suitable construction.	Components are witnessed at normal condition and abnormal test are verified	P
9.1.1	Reducing the risk of ignition and spread of flame		P
	For equipment or a portion of equipment, there are two alternative methods of providing protection against ignition and spread of flame that could affect materials, wiring, wound components and electronic components such as integrated circuits, transistors, thyristors, diodes, resistors and capacitors.	Method 1 used	P
9.1.2	Conditions for a fire enclosure		P
	A FIRE ENCLOSURE is required for equipment or parts of equipment for which Method 2 is not fully applied and complied with.		P
9.1.2.1	Parts requiring a fire enclosure		P
	Except where Method 2 is used, or as permitted in 9.1.2.2, the following are considered to have a risk of ignition and, therefore, require a FIRE ENCLOSURE:		P
	– components in PRIMARY CIRCUITS		P
	– components in SECONDARY CIRCUITS supplied by power sources which exceed the limits for a		P
	LIMITED POWER SOURCE as specified in 9.2;		
	– components in SECONDARY CIRCUITS supplied by a LIMITED POWER SOURCE as specified in 9.2, but not mounted on a material of FLAMMABILITY CLASS V-1;	PWB rated V-0	N/A
	– components within a power supply unit or assembly having a limited power output complying with the criteria for a LIMITED POWER SOURCE as specified in 9.2, including overcurrent protective devices, limiting impedances, regulating networks and wiring, up to the point where the LIMITED POWER SOURCE output criteria are met;		P
	– components having unenclosed arcing parts, such as open switch and relay contacts and commutators, in a circuit at HAZARDOUS VOLTAGE or at a HAZARDOUS ENERGY LEVEL; and		N/A
	– insulated wiring, except as permitted in 9.1.2.2.		N/A
9.1.2.2	Parts not requiring a fire enclosure		N/A

9.1.3	Materials requirements for protection against fire hazard		P
9.1.3.1	General		P
	ENCLOSURES, components and other parts shall be so constructed, or shall make use of such materials, that the propagation of fire is limited.		P
9.1.3.2	Materials for fire enclosures		P
	If an enclosure material is not classified as specified below, a test may be performed on the final enclosure or part of the enclosure, in which case the material shall additionally be subjected to periodic SAMPLE testing.		P
9.1.3.3	Materials for components and other parts outside fire enclosures		P
	Except as otherwise noted below, materials for components and other parts (including MECHANICAL ENCLOSURES, ELECTRICAL ENCLOSURES and DECORATIVE PARTS); located outside FIRE ENCLOSURES, shall be of FLAMMABILITY CLASS HB.	Internal wire:VW-1 PWB: V-0	P
9.1.3.4	Materials for components and other parts inside fire enclosures		N/A
9.1.3.5	Materials for air filter assemblies		N/A
9.1.4	Openings in fire enclosures		N/A
9.1.4.1	General		N/A
	For equipment that is intended to be used or installed in more than one orientation as specified in the product documentation, the following requirements apply in each orientation.		N/A
	following sections:		
	– 7.3.4, Protection against direct contact;		N/A
	– 7.4, Protection against energy hazards;		N/A
	– 13.5, Openings in enclosures		N/A
9.1.4.2	Side openings treated as bottom openings		N/A
9.1.4.3	Openings in the bottom of a fire enclosure		N/A
	The bottom of a FIRE ENCLOSURE or individual barriers, shall provide protection against emission of flaming or molten material under all internal parts, including partially enclosed components or assemblies, for which Method 2 of 9.1.1 has not been fully applied and complied with.		N/A
9.1.4.4	Equipment for use in a CLOSED ELECTRICAL OPERATING AREA		N/A

	The requirements of 9.1.4.3 do not apply to FIXED EQUIPMENT intended only for use in a CLOSED ELECTRICAL OPERATING AREA and to be mounted on a concrete floor or other non- combustible surface. Such equipment shall be marked as follows:		N/A
	WARNING: FIRE HAZARD SUITABLE FOR MOUNTING ON CONCRETE OR OTHER NON- COMBUSTIBLE SURFACE ONLY		N/A
9.1.4.5	Doors or covers in fire enclosures		N/A
9.1.4.6	Additional requirements for openings in transportable equipment		N/A
9.2	LIMITED POWER SOURCES		N/A
9.2.1	General		N/A
9.2.2	Limited power source tests		N/A
9.3	Short-circuit and overcurrent protection		P
9.3.1	General		P
	The PCE shall not present a hazard, under short- circuit or overcurrent conditions at any port, including phase-to-phase, phase-to-earth and phase-to-neutral, and adequate information shall be provided to allow proper selection of external wiring and external protective devices.		P
9.3.2	Protection against short-circuits and overcurrents shall be provided for all input circuits, and for output circuits that do not comply with the requirements for limited power sources in 9.2, except for circuits in which no overcurrent hazard is presented by short- circuits and overloads.	Fuse integral to PCE	P
9.3.3	Protective devices provided or specified shall have adequate breaking capacity to interrupt the maximum short circuit current specified for the port to which they are connected. If protection that is provided integral to the PCE for an input port is not rated for the short-circuit current of the circuit in	Fuse integral to PCE	P
	These requirements are in addition to those in the		N/A
	which it is used, the installation instructions shall specify that an upstream protective device, rated for the prospective short-circuit current of that port, shall be used to provide backup protection.		
10	PROTECTION AGAINST SONIC PRESSURE HAZARDS		P
10.1	General		P

	The equipment shall provide protection against the effect of sonic pressure. Conformity tests are carried out if the equipment is likely to cause such HAZARDS.	No sonic pressure hazards.	P
10.2	Sonic pressure and Sound level	< 85dB	P
10.2.1	Hazardous Noise Levels		P
11	PROTECTION AGAINST LIQUID HAZARDS		N/A
11.1	Liquid Containment, Pressure and Leakage	No liquid containment system	N/A
	The liquid containment system components shall be compatible with the liquid to be used.		N/A
	There shall be no leakage of liquid onto live parts as a result of:		N/A
	a) Normal operation, including condensation;		N/A
	b) Servicing of the equipment; or		N/A
	c) Inadvertent loosening or detachment of hoses or other cooling system parts over time.		N/A
11.2	Fluid pressure and leakage		N/A
11.2.1	Maximum pressure		N/A
11.2.2	Leakage from parts		N/A
11.2.3	Overpressure safety device		N/A
11.3	Oil and grease		N/A
12	CHEMICAL HAZARDS		N/A
12.1	General		N/A
13	PHYSICAL REQUIREMENTS		P
13.1	Handles and manual controls		N/A
	Handles, knobs, grips, levers and the like shall be reliably fixed so that they will not work loose in normal use, if this might result in a hazard. Sealing compounds and the like, other than self-hardening resins, shall not be used to prevent loosening. If handles, knobs and the like are used to indicate the position of switches or similar components, it shall not be possible to fix them in a wrong position if this might result in hazard.		N/A
13.1.1	Adjustable controls		N/A
13.2	Securing of parts		P
13.3	Provisions for external connections		P
13.3.1	General		--
13.3.2	Connection to an a.c. Mains supply		P
13.3.2.1	General		P
	For safe and reliable connection to a MAINS supply, equipment shall be provided with one of the following:		P

	– terminals or leads or a non-detachable power supply cord for permanent connection to the supply; or		P
	– a non-detachable power supply cord for connection to the supply by means of a plug		N/A
	– an appliance inlet for connection of a detachable power supply cord; or		N/A
	– a mains plug that is part of direct plug-in equipment as in 13.3.8		N/A
13.3.2.2	Permanently connected equipment		P
13.3.2.3	Appliance inlets		P
13.3.2.4	Power supply cord		N/A
13.3.2.5	Cord anchorages and strain relief		N/A
	For equipment with a non-detachable power supply cord, a cord anchorage shall be supplied such that:		N/A
	– the connecting points of the cord conductors are relieved from strain; and		N/A
	– the outer covering of the cord is protected from abrasion.		N/A
13.3.2.6	Protection against mechanical damage	Metal enclosure provided	P
13.3.3	Wiring terminals for connection of external conductors		P
13.3.3.1	Wiring terminals		P
13.3.3.2	Screw terminals		P
13.3.3.3	Wiring terminal sizes		P
13.3.3.4	Wiring terminal design		P
13.3.3.5	Grouping of wiring terminals		P
13.3.3.6	Stranded wire		N/A
13.3.4	Supply wiring space		N/A
13.3.5	Wire bending space for wires 10 mm ² and greater		N/A
13.3.6	Disconnection from supply sources	User manual mentioned	P
13.3.7	Connectors, plugs and sockets		P
13.3.8	Direct plug-in equipment		N/A
13.4	Internal wiring and connections		P
13.4.1	General		P
13.4.2	Routing		P
13.4.3	Colour coding	Green-yellow wire used as protective bonding only	P
13.4.4	Splices and connections		P
13.4.5	Interconnections between parts of the PCE		P
13.5	Openings in enclosures		N/A
13.5.1	Top and side openings		N/A

	Openings in the top and sides of ENCLOSURES shall be so located or constructed that it is unlikely that objects will enter the openings and create hazards by contacting bare conductive parts.		N/A
13.6	Polymeric Materials		P
13.6.1	General		P
13.6.1.1	Thermal index or capability		P
13.6.2	Polymers serving as enclosures or barriers preventing access to hazards		P
13.6.2.1	Stress relief test		N/A
13.6.3	Polymers serving as solid insulation		P
13.6.3.1	Resistance to arcing		N/A
13.6.4	UV resistance		P
	Polymeric parts of an OUTDOOR ENCLOSURE required for compliance with this standard shall be sufficiently resistance to degradation by ultra-violet (UV) radiation	The enclosure of the unit is made of metal with painting.	P
13.7	Mechanical resistance to deflection, impact, or drop		P
13.7.1	General		P
13.7.2	250-N deflection test for metal enclosures		P
13.7.3	7-J impact test for polymeric enclosures		P
13.7.4	Drop test		N/A
13.8	Thickness requirements for metal enclosures		P
13.8.1	General		P
13.8.2	Cast metal		N/A
13.8.3	Sheet metal	> 0.8mm	P
14	COMPONENTS		P
14.1	General	(see appended table 14)	P
	Where safety is involved, components shall be used in accordance with their specified RATINGS unless a specific exception is made. They shall conform to one of the following:		P
	a) applicable safety requirements of a relevant IEC standard. Conformity with other requirements of the component standard is not required. If necessary for the application, components shall be subjected to the test of this standard, except that it is not necessary to carry out identical or equivalent tests already performed to check conformity with the component standard;		P
	b) the requirements of this standard and, where		P
	necessary for the application, any additional applicable safety requirements of the relevant IEC component standard;		

	c) if there is no relevant IEC standard, the requirements of this standard;		P
	d) applicable safety requirements of a non-IEC standard which are at least as high as those of the applicable IEC standard, provided that the component has been approved to the non-IEC standard by a recognized testing authority.		P
	Components such as optocouplers, capacitors, transformers, and relays connected across basic, supplemental, reinforced, or double insulation shall comply with the requirements applicable for the grade of insulation being bridged, and if not previously certified to the applicable component safety standard shall be subjected to the voltage test of 7.5.2 as routine test.		P
14.2	Motor Over Temperature Protection		N/A
	Motors which, when stopped or prevented from starting (see 4.4.4.3), would present an electric shock HAZARD, a temperature HAZARD, or a fire HAZARD, shall be protected by an over temperature or thermal protection device meeting the requirements of 14.3.	No motor	N/A
14.3	Over temperature protection devices		P
14.4	Fuse holders		N/A
14.5	MAINS voltage selecting devices		N/A
14.6	Printed circuit boards		P
	Printed circuit boards shall be made of material with a flammability classification of V-1 of IEC 60707 or better.	V-0	P
	This requirement does not apply to thin-film flexible printed circuit boards that contain only circuits powered from limited power sources meeting the requirements of 9.2.		P
	Conformity of the flammability RATING is checked by inspection of data on the materials. Alternatively, conformity is checked by performing the V-1 tests specified in IEC 60707 on three samples of the relevant parts.		P
14.7	Circuits or components used as transient overvoltage limiting devices		N/A
	If control of transient overvoltage is employed in the equipment, any overvoltage limiting component or circuit shall be tested with the applicable impulse withstand voltage of Table 7-10 using the test method from 7.5.1 except 10 positive and 10 negative impulses are to be applied and may be spaced up to 1 min apart.		N/A
14.8	Batteries		N/A

	Equipment containing batteries shall be designed to reduce the risk of fire, explosion and chemical leaks under normal conditions and after a single fault in the equipment including a fault in circuitry within the equipment battery pack.		N/A
14.8.1	Battery Enclosure Ventilation		N/A
14.8.1.1	Ventilation requirements		N/A
14.8.1.2	Ventilation testing		N/A
14.8.1.3	Ventilation instructions		N/A
14.8.2	Battery Mounting		N/A
	Compliance is verified by the application of the force to the battery's mounting surface. The test force is to be increased gradually so as to reach the required value in 5 to 10 s, and is to be maintained at that value for 1 min. A non-metallic rack or tray shall be tested at the highest normal condition operating temperature.		N/A
14.8.3	Electrolyte spillage		N/A
	Battery trays and cabinets shall have an electrolyte-resistant coating.		N/A
	The ENCLOSURE or compartment housing a VENTED BATTERY shall be constructed so that spillage or leakage of the electrolyte from one battery will be contained within the ENCLOSURE and be prevented from:		N/A
	a) reaching the PCE outer surfaces that can be contacted by the USER		N/A
	b) contaminating adjacent electrical components or materials; and		N/A
	c) bridging required electrical distances		N/A
14.8.4	Battery Connections		N/A
	Reverse battery connection of the terminals shall be prevented if reverse connection could result in a hazard within the meaning of this Standard		N/A
14.8.5	Battery maintenance instructions		N/A
	The information and instructions listed in 5.3.4.1 shall be included in the operator manual for equipment in which battery maintenance is performed by the operator, or in the service manual if battery maintenance is to be performed by service personnel only.		N/A
14.8.6	Battery accessibility and maintainability		N/A

	Battery terminals and connectors shall be accessible for maintenance with the correct TOOLS. Batteries with liquid electrolyte, requiring maintained shall be so located that the battery cell caps are accessible for electrolyte tests and readjusting of electrolyte levels.		N/A
15	Software and firmware performing safety functions		P
Annex A	Measurement of clearances and creepage distances (see 7.3.7.4 and 7.3.7.5)		P
Annex B	Programmable Equipment		P
B.1	Software or firmware that perform safety critical functions		P
B.1.1	Firmware or software that performs a critical safety function/s, the failure of which can result in a risk of fire, electric shock or other hazard as specified by this standard, shall be evaluated by one of the following means.		P
	a) All software or firmware limits or controls shall be disabled before the test to evaluate the hardware circuitry during the abnormal test condition related to the safety function.		P
	b) Protective controls employing software or firmware to perform their function(s), shall be so constructed that they comply with IEC 60730-1 Annex H to address the risks identified in B.2.1.		N/A
B.2	Evaluation of controls employing software		P
Annex C	Symbols to be used in equipment markings		P
Annex D	Test Probes for Determining Access		P
Annex E	RCDs		N/A
Annex F	Altitude correction for clearances		N/A
Annex G	Clearance and creepage distance determination for frequencies greater than 30 kHz		P
Annex H	Measuring Instrument for Touch Current Measurements		P
H.1	Measuring instrument		P
H.2	Alternative measuring instrument		N/A
Annex I	Examples of Protection, Insulation, and Overvoltage Category Requirements for PCE		P
Annex J	Ultraviolet light conditioning test	LCD cover is approved by UL according to UV resistance.	P

4.7	Electrical ratings tests						P
Input Voltage (V)	Input Current (A)	Input Power (kW)	Output Voltage (V)	Output Current (A)	Output power (kW)	F (HZ)	
180Vd.c.	30.27	5.449	220Va.c.	19.92	4.382	49.97	
180Vd.c.	30.20	5.436	220Va.c.	19.85	4.367	49.97	
180Vd.c.	30.22	5.440	220Va.c.	19.57	4.305	49.97	
450Vd.c.	12.22	5.499	220Va.c.	19.92	4.382	49.97	
450Vd.c.	12.18	5.481	220Va.c.	19.89	4.376	49.97	
450Vd.c.	12.12	5.454	220Va.c.	19.78	4.352	49.97	
Supplementary information: N/A							

4.3	TABLE: heating temperature rise measurements				P	
	test voltage (V)	Input: 450Vdc Output: 220Vac		—		
	t1 (°C)	50.0		--		
	t2 (°C)	50.0		--		
Thermocouple Locations		Max. temperature measured (°C)		Max. temperature limit (°C)		
Ambient		52.00		--		
PV terminal		55.81		105		
input terminal		51.94		105		
output terminal		50.72		105		
PCB near copper foil		69.80		105		
C24		58.46		105		
F1		57.52		130		
U1		58.02		110		
C10		56.75		105		
C29		56.98		105		
R1		57.48		105		
Q2		65.05		175		
U3		50.75		85		
U4		54.62		85		
Side surface of enclosure		51.45		60		
Top surface of enclosure		49.93		60		
Front surface of enclosure		49.76		60		
Back surface of enclosure		49.55		60		
	TABLE: Heating test, resistance method					
	Test voltage (V)..... :			--		
	Ambient, t1 (°C)			--		
	Ambient, t2 (°C)			--		
Temperature rise of winding		R1 (Ω)	R2 (Ω)	$\Delta T(K)$	Max. dT (K)	Insulation
Supplementary information: N/A						

4.4	TABLE: fault condition tests						P
		ambient temperature (°C) : 25				--	
No.	component No.	fault	test voltage (V)	test time	Fuse No.	Fuse current (A)	result
1	Rs	S/C	450	10min	--	--	Circuit protected, recoverable.
2	R9	O/C	450	10min	--	--	Circuit protected immediately, no output, recoverable.
3	C17	S/C	450	10min	--	--	Output voltage up to 470V, No hazard
4	C31	S/C	450	10min	--	--	Circuit protected immediately, no hazard, coverable.
5	C12	S/C	450	10min	--	--	No charging current, no hazard, recoverable.
6	C20	S/C	450	10min	--	--	No hazard, recoverable.
7	C18	S/C	450	10min	--	--	Fuse broken, no hazard.
8	Q2(D-S)	S/C	450	10min			Fuse broken, no hazard.
supplementary information							
S/C: Short circuit, O/C: Open circuit During the test:							
Fire do not propagate beyond the EUT; Equipment do not emitt molten metal;							
Enclosures do not deform to cause non-compliance with the standard. Pass the dielectric test.							

7.3.7	TABLE: clearance and creepage distance measurements						P
clearance cl and creepage distance dcr at / of:		Up (V)	U r.m.s. (V)	required cl (mm)	cl (mm)	required dcr (mm)	dcr (mm)
Between live parts and isolated interface(RI)		150	450	1.5*1.48=2.1	2.5	3.2*1.48=4.74	5.7
Between live parts on PCB and enclosure(BI)		150	450	1.5*1.48=2.1	5.0	1.6*1.48=2.36	5.0

7.3.7.8.	TABLE: distance through insulation measurement						P
3.2 to 7.3.7.8.							
distance through insulation di at/of:				U r.m.s. (V)	test voltage	required dti (mm)	dti (mm)
Communication isolated optocouplers (BI)				1000 Vd.c. 277 Va.c.	1680 Vr.m.s	>0,4	Certified

Multi-layer insulated winding wire as sec. winding in iso- lated transformer (RI)	1000 Vd.c. 277 Va.c.	1680 Vr.m.s	--	Certified
Note: detail see Critical components list (CDF)				

7.5	TABLE: electric strength measurements, impulse voltage test and partial discharge test				P
test voltage applied between:		test voltage (V)	Impulse withstand voltage (V)	partial discharge extinction voltage (V)	result
PV input and Enclosure		450Vd.c	1500Vpeak	N/A	No breakdown
input and Enclosure (Connect directly to mains)		450Vd.c	4000Vpeak	N/A	No breakdown
output side and Enclosure		220Va.c	4000Vpeak	N/A	No breakdown
PV input and Communication port		450Vd.c	2500Vpeak	N/A	No breakdown
input and Communication port (Connect directly to mains)		450Vd.c	4000Vpeak	N/A	No breakdown
output side and Communication port		220Va.c	4000Vpeak	N/A	No breakdown

14	TABLE: list of critical components					P
object/part No.	manufacturer / trademark	type/model	technical data	standard	mark(s) of conformity1)	
Plastic enclosure	Zhejiang Yongxin	5.5KW B503SL-2007-SS2Single-phase	V-0, 105°C	UL 94	UL	
PCB	Guangde Infield	3030 drive plate	ABS, V-0, 130 °C	UL 796	UL	
Internal wire	Jiaxing Duojiang	BV1.5m2	1.5m2 Round aluminum wire	GB/T 5023.3-2008/IEC60227-3:1997	3C VDE	
Fan	Shenzhen Yingtian	YTD248025B	80*80*25mm,3200RMP,24Vdc	IEC 62109-1, IEC 62109-2	Tested with appliance	
Heat sink	Shanghai Lianglu	LX-SRQ3030	Aluminum products	-	ROHS	
Plastic cover for protective LCD	Cangzhou Caijia	LX320E	Max.50VDC	-	ROHS	
PV connector	Fuanlin	BA8-11-14.5-00-C	600V, 40A	-	ROHS	

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4	General testing requirements		P
4.4.4	Single fault conditions to be applied		P
4.4.4.15	Fault-tolerance of protection for grid-interactive INVERTERs		P
4.4.4.15.1	Fault-tolerance of residual current monitoring according to 4.8.3.5: the residual current monitoring system operates properly	(see appended table) 4.4.4.15.1	P
	a). - The INVERTER ceases to operate		P
	- Indicates a fault in accordance with §13.9		P
	- Disconnect from the mains		P
	- not re-connect after any sequence of removing and reconnecting PV power		P
	- not re-connect after any sequence of removing and reconnecting AC power		P
	- not re-connect after any sequence of removing and reconnecting both PV and AC power		P
	b). - The INVERTER continues to operate		N/A
	- the residual current monitoring system operates properly under single fault condition		N/A
	- Indicates a fault in accordance with §13.9		N/A
	c) . - The INVERTER continues to operate regardless of loss of residual current monitoring functionality		N/A
	- not re-connect after any sequence of removing and reconnecting PV power		N/A
	- not re-connect after any sequence of removing and reconnecting AC power		N/A
	- not re-connect after any sequence of removing and reconnecting both PV and AC power		N/A
	- Indicates a fault in accordance with §13.9		N/A
4.4.4.15.2	Fault-tolerance of automatic disconnecting means		P
4.4.4.15.2.1	The means provided for automatic disconnection of a grid-interactive INVERTER from the mains shall:		P
	- disconnect all grounded current-carrying conductors from the mains		P
	- disconnect all ungrounded current-carrying conductors from the mains		P
	- be such that with a single fault applied to the disconnection means or to any other location in the INVERTER, at least basic insulation or simple separation is maintained between the PV array and the mains when the disconnecting means is intended to be in the open state.	(see appended table) 4.4.4.15.2 Fault-tolerance of automatic disconnecting	P

4.4.4.15.2.2	Design of insulation or separation complies with requirements of 7.3.7 of Part 1: report here Part 1 comment and verdict.		P
4.4.4.15.2.3	For non-isolated INVERTER, automatic checking of the isolation provided by a disconnect means after single fault.	(see appended table) 4.4.4.15.2 Fault-tolerance of automatic disconnecting.	P
	If the check fail: any still-functional disconnection means shall be left in the open position		P
	- at least basic or simple separation shall be maintained between the PV input and the mains		P
	- the INVERTER shall not start operation		P
	- the INVERTER shall indicate a fault in accordance with 13.9		P
4.4.4.16	A stand-alone INVERTER with a transfer switch to transfer AC loads from the mains or other AC bypass source to the INVERTER output:	Grid interactive INVERTER.	N/A
	- shall continue to operate normally		N/A
	- shall not present a risk of fire as the result of an out-of-phase transfer		N/A
	- shall not present a risk of shock as the result of an out-of-phase transfer		N/A
	- And having control preventing switching: components for malfunctioning		N/A
4.4.4.17	Cooling system failure – Blanketing test No hazards according to the criteria of sub-clause 4.4.3 of Part 1 shall result from blanketing the INVERTER. This test is not required for INVERTERs restricted to use only in closed electrical operating areas.	(see appended table) Cooling system failure – Blanketing test.	P
	Test stop condition: time duration value or stabilized temperature		P
4.7	Electrical ratings tests		N/A
4.7.4	Stand-alone INVERTER AC output voltage and frequency		N/A
4.7.4.1	General		N/A
4.7.4.2	Steady state output voltage at nominal DC input The steady-state AC output voltage shall not be less than 90 % or more than 110 % of the rated nominal voltage with the INVERTER supplied with its nominal value of DC input voltage.		N/A
4.7.4.3	Steady state output voltage across the DC input Range The steady-state AC output voltage shall not be less than 85 % or more than 110 % of the rated nominal voltage with the INVERTER supplied with any value within the rated		N/A

	range of DC input voltage.		
4.7.4.4	Load step response of the output voltage at nominal DC input The AC output voltage shall not be less than 85 % or more than 110 % of the rated nominal voltage for more than 1,5 s after application or removal of a resistive load.		N/A
4.7.4.5	Steady state output frequency The steady-state AC output frequency shall not		N/A
	vary from the nominal value by more than +4 % or -6 %.		
4.7.5	Stand-alone INVERTER output voltage waveform		N/A
4.7.5.1	General		N/A
4.7.5.2	The AC output voltage waveform of a sinusoidal output stand-alone INVERTER shall have a total harmonic distortion (THD) not exceeding of 10 % and no individual harmonic at a level exceeding 6%.		
4.7.5.3	Non-sinusoidal output waveform requirements		N/A
4.7.5.3.1	General		N/A
4.7.5.3.2	The total harmonic distortion (THD) of the voltage waveform shall not exceed 40 %.		N/A
4.7.5.3.3	The slope of the rising and falling edges of the positive and negative half-cycles of the voltage waveform shall not exceed 10 V/ μ s measured between the points at which the waveform has a voltage of 10 % and 90 % of the peak voltage for that half-cycle.		N/A
4.7.5.3.4	The absolute value of the peak voltage of the positive and negative half-cycles of the waveform shall not exceed 1,414 times 110 % of the RMS value of the rated nominal AC output voltage.		N/A
4.7.5.4	Information requirements for non-sinusoidal waveforms The instructions provided with a stand- alone INVERTER not complying with 4.7.5.2 shall include the information in 5.3.2.6.		N/A
4.7.5.5	Output voltage waveform requirements for INVERTERs for dedicated loads. For an INVERTER that is intended only for use with a known dedicated load, the following requirements may be used as an alternative to the waveform requirements in 4.7.5.2 to 4.7.5.3.		N/A
	The combination of the INVERTER and dedicated load shall be evaluated to ensure that the output waveform does not cause any hazards in the load equipment and INVERTER, or cause the load equipment to fail to comply		N/A

	with the applicable product safety standards.		
	The INVERTER shall be marked with symbols 9 and 15 of Table C.1 of Part 1.		N/A
	The installation instructions provided with the INVERTER shall include the information in 5.3.2.13.		N/A
4.8	Additional tests for grid-interactive INVERTERs		P
4.8.1	General requirements regarding INVERTER isolation and array grounding		N/A
	- Type of Array grounding supported:		N/A
	- INVERTER isolation:		N/A
4.8.2	Array insulation resistance detection for INVERTERs for ungrounded and functionally grounded arrays	(see appended table)	P
4.8.2.1	Array insulation resistance detection for INVERTERs for ungrounded arrays		P
	INVERTER shall have means to measure DC insulation resistance from PV input (array) to ground before starting operation		P
	Or INVERTER shall be provided with instruction in accordance with 5.3.2.11.		N/A
	Measured DC insulation resistance:		P
	INVERTER measurement circuit shall be capable of detecting insulation resistance below the limit value $R = V_{max}/30mA$ under normal conditions		P
	INVERTER measurement circuit shall be capable of detecting insulation resistance below the limit value $R = V_{max}/30mA$ with ground fault in the PV array		P
	Isolated INVERTERs shall indicate a fault if the insulation resistance is less than the limit value		N/A
	Isolated INVERTER fault indication maintained until insulation resistance has recovered to a value higher than the limit value		N/A
	Non-isolated INVERTERs, or INVERTERs with isolation not complying with the leakage current limits in the minimum INVERTER isolation requirements in Table 30:		P
	- shall indicate a fault in accordance with 13.9		P
	- shall not connect to the mains		P
4.8.2.2	Array insulation resistance detection for INVERTERs for functionally grounded arrays		N/A

	a-1)The value of the total resistance, including the intentional resistance for array functional grounding, the expected insulation resistance of the array to ground, and the resistance of any other networks connected to ground (for example measurement networks) must not be lower than $R = (V_{MAX} PV/30 \text{ mA}) \text{ ohms}$.		N/A
	a-2) The installation instructions shall include the information required in 5.3.2.12.		N/A
	b-1) As an alternative to a), or if a resistor value lower than in a) is used, the INVERTER shall incorporate means to detect, during operation, if the total current through the resistor and any networks (for example measurement networks) in parallel with it, exceeds the residual current values and times in Table 31		N/A
	b-2) INVERTER shall either disconnect the resistor or limit the current by other means		N/A
	b-3) If the INVERTER is a non-isolated INVERTER, or has isolation not complying with the leakage current limits in the minimum INVERTER isolation requirements in Table 30, it shall also disconnect from the mains.		N/A
	c) The INVERTER shall have means to measure the DC insulation resistance from the PV input to ground before starting operation, in accordance with 4.8.2.1.		N/A
4.8.3	Array residual current detection		P
4.8.3.1	General		P
4.8.3.2	30 mA touch current type test for isolated INVERTERs		N/A
4.8.3.3	Fire hazard residual current type test for isolated INVERTERs		N/A
4.8.3.4	Protection by application of RCD's		N/A
	- The requirement for additional protection in 4.8.3.1 can be met by provision of an RCD with a residual current setting of 30 mA, located between the INVERTER and the mains..		N/A
	- The selection of the RCD type to ensure compatibility with the INVERTER must be made according to rules for RCD selection in Part 1.		N/A
	- The RCD provided integral to the INVERTER, or		N/A
	- The RDC provided by the installer if details of the rating, type, and location for the RCD are given in the installation instructions per 5.3.2.9.		N/A
4.8.3.5	Protection by residual current monitoring		P
4.8.3.5.1	General		P

	Where required by Table 30, the INVERTER shall provide residual current monitoring that functions whenever the INVERTER is connected to the mains with the automatic disconnection means closed.		P
	The residual current monitoring means shall measure the total (both a.c. and d.c. components) RMS current.		P
	As indicated in Table 30 for different INVERTER types, array types, and INVERTER isolation levels, detection may be required for excessive continuous residual current, excessive sudden changes in residual current, or both, according to the following limits:		P
	a) Continuous residual current: The INVERTER shall disconnect within 0,3 s and indicate a fault in accordance with 13.9 if the continuous residual current exceeds:		P
	- maximum 300 mA for INVERTERs with continuous output power rating ≤ 30 kV;		N/A
	- maximum 10 mA per kVA of rated continuous output power for INVERTERs with continuous output power rating > 30 kVA.		P
	The INVERTER may attempt to re-connect if the array insulation resistance meets the limit in 4.8.2.		N/A
	b) Sudden changes in residual current: The INVERTER shall disconnect from the mains within the time specified in Table 31		P
	The INVERTER indicates a fault in accordance with 13.9, if a sudden increase in the RMS residual		P
	current is detected exceeding the value in the table.		
	The INVERTER may attempt to re-connect if the array insulation resistance meets the limit in 4.8.2.		N/A
4.8.3.5.2	Test for detection of excessive continuous residual current: test repeated 5 times and time to disconnect shall not exceed 0,3 s.	(see appended table) 4.8.3.5.2 Test for detection of excessive continuous residual current.	P
4.8.3.5.3	Test for detection of sudden changes in residual current repeated 5 times and each of the 5 results shall not exceed the time limit indicated in for each row (30mA, 60mA and 150mA) of Table 31.	(see appended table)	P
4.8.3.6	Systems located in closed electrical operating areas		N/A
	The protection against shock hazard is not required if the installation information provided with the INVERTER indicates the restriction for use in a closed electrical operating area, and		N/A

	Installation information indicates what forms of shock hazard protection are and are not provided integral to the INVERTER, in accordance with 5.3.2.7.		N/A
	The INVERTER shall be marked as in 5.2.2.6.		N/A
5	MARKING AND DOCUMENTATION		P
5.1	Marking		P
5.1.4	Equipment ratings		P
	PV input ratings:		P
	- Vmax PV (absolute maximum) (d.c. V)		P
	- Isc PV (absolute maximum) (d.c. A)		P
	a.c. output ratings:		P
	- Voltage (nominal or range) (a.c. V)		P
	- Current (maximum continuous) (a.c. A)		P
	- Frequency (nominal or range) (Hz)		P
	- Power (maximum continuous) (W or VA)		P
	- Power factor range		P
	a.c input ratings:		N/A
	- Voltage (nominal or range) (a.c. V)		N/A
	- Current (maximum continuous) (a.c. A)		N/A
	- Frequency (nominal or range) (Hz)		N/A
	d.c. output ratings:		N/A
	- Voltage (nominal or range) (d.c. V)		N/A
	- Current (maximum continuous) (d.c. A)		N/A
	Protective class (I or II or III)		P
	Ingress protection (IP) rating per part 1		P
	An INVERTER that is adjustable for more than one		N/A
	nominal output voltage shall be marked to indicate the particular voltage for which it is set when shipped from the factory.		
5.2	Warning markings		N/A
5.2.2	Content for warning markings		N/A
5.2.2.6	INVERTERs for closed electrical operating areas		N/A
	Where required by 4.8.3.6, an INVERTER not provided with full protection against shock hazard on the PV array shall be marked with a warning that the INVERTER is only for use in a closed electrical operating area, and referring to the installation instructions.		N/A
5.3	Documentation		P
5.3.2	Information related to installation		P

5.3.2.1	Ratings. Subclause 5.3.2 of Part 1 requires the documentation to include ratings information for each input and output. For INVERTERs this information shall be as in Table 33 below. Only those ratings that are applicable based on the type of INVERTER are required.	P
	PV input quantities :	P
	- Vmax PV (absolute maximum) (d.c. V)	P
	- PV input operating voltage range (d.c. V)	P
	- Maximum operating PV input current (d.c. A)	P
	- Isc PV (absolute maximum) (d.c. A)	P
	- Max. INVERTER backfeed current to the array (a.c. or d.c. A)	P
	a.c. output quantities:	P
	- Voltage (nominal or range) (a.c. V)	P
	- Current (maximum continuous) (a.c. A)	P
	- Current (inrush) (a.c. A, peak and duration)	P
	- Frequency (nominal or range) (Hz)	P
	- Power (maximum continuous) (W or VA)	P
	- Power factor range	P
	- Maximum output fault current (a.c. A, peak and duration or RMS)	P
	- Maximum output overcurrent protection (a.c. A)	P
	a.c. input quantities:	N/A
	- Voltage (nominal or range) (a.c. V)	N/A
	- Current (maximum continuous) (a.c. A)	N/A
	- Current (inrush) (a.c. A, peak and duration)	N/A
	- Frequency (nominal or range) (Hz)	N/A
	d.c input (other than PV) quantities:	N/A
	- Voltage (nominal or range) (d.c. V)	N/A
	- Nominal battery voltage (d.c. V)	N/A
	- Current (maximum continuous) (d.c. A)	N/A
	d.c. output quantities:	N/A
	- Voltage (nominal or range) (d.c. V)	N/A
	- Nominal battery voltage (d.c. V)	N/A
	- Current (maximum continuous) (d.c. A)	N/A
	Protective class (I or II or III)	P
	Ingress protection (IP) rating per part 1	P
5.3.2.2	Grid-interactive INVERTER setpoints	Non-adjustable to operator, pre-set by manufacturer before shipment.

	For a grid-interactive unit with field adjustable trip points, trip times, or reconnect times, the presence of such controls, the means for adjustment, the factory default values, and the limits of the ranges of adjustability shall be provided in the documentation for the PCE or in other format such as on a website. Provided solution:		N/A
	The setting of field adjustable setpoints shall be accessible from the PCE		N/A
5.3.2.3	Transformers and isolation		P
	whether an internal isolation transformer is provided, and if so, what level of insulation (functional, basic, reinforced, or double) is provided by that transformer. The instructions shall also indicate what the resulting installation requirements are regarding such things as earthing or not earthing the array, providing external residual current detection devices, etc.	The PV array shall not be earthed. External isolate transformer shall used for models: 7.5KW LX100-7R5G-4-PV	P
	An INVERTER shall be provided with information to the installer regarding:		P
	- providing of internal isolation transformer		N/A
	- the level of insulation (functional, basic, reinforced, or double)	External isolate transformer shall used for models: 7.5KW LX100-7R5G-4-PV	P
	The instructions shall also indicate what the resulting installation requirements are regarding:		P
	- earthing or not earthing the array	Not earthing	P
	- providing external residual current detection devices		N/A
	- requiring an external isolation transformer,	Isolated frequency transformer.	P
5.3.2.4	Transformers required but not provided		P
	An INVERTER that requires an external isolation transformer not provided with the unit, shall be provided with instructions that specify, and for the external isolation transformer with which it is intended to be used:		P
	- the configuration type		P
	- electrical ratings		P
	- environmental ratings		P
5.3.2.5	PV modules for non-isolated INVERTERS		P
	Non-isolated INVERTERS shall be provided with installation instructions that require PV modules that have an IEC 61730 Class A rating		P
	If the maximum AC mains operating voltage is higher than the PV array maximum system voltage then the instructions shall require PV modules that have a maximum system voltage rating based upon the AC mains voltage.		N/A

5.3.2.6	Non-sinusoidal output waveform information		N/A
	The instruction manual for a stand-alone INVERTER not complying with 4.7.5.2 shall include a warning that:		N/A
	- the waveform is not sinusoidal,		N/A
	- some loads may experience increased heating,		N/A
	- the user should consult the manufacturers of the intended load equipment before operating that load with the INVERTER		N/A
	The INVERTER manufacturer shall provide information regarding:		N/A
	- what types of loads may experience increased heating		N/A
	- recommendations for maximum operating times with such loads		N/A
	The INVERTER manufacturer shall specify for the waveforms as determined by the testing in 4.7.5.3.2 through 4.7.5.3.4.:		N/A
	- THD		N/A
	- slope		N/A
	- peak voltage		N/A
5.3.2.7	Systems located in closed electrical operating areas		N/A
	Where required by 4.8.3.6, an INVERTER not provided with full protection against shock hazard on the PV array shall be provided with installation instructions:		N/A
	- requiring that the INVERTER and the array must be installed in closed electrical operating areas		N/A
	- indicating which forms of shock hazard protection are and are not provided integral to the INVERTER (for example the RCD, isolation transformer complying with the 30 mA touch current limit, or residual current monitoring for sudden changes)		N/A
5.3.2.8	Stand-alone INVERTER output circuit bonding		N/A
	Where required by 7.3.10, the documentation for an INVERTER shall include the following:		N/A
	- if output circuit bonding is required but is not provided integral to the INVERTER, the required means shall be described in the installation instructions, including which conductor is to be bonded and the required current carrying capability		N/A
	or cross-section of the bonding means;		
	- if the output circuit is intended to be floating, the documentation for the INVERTER shall indicate that the output is floating.		N/A
5.3.2.9	Protection by application of RCD's		N/A

	Where the requirement for additional protection in 4.8.3.1 is met by requiring an RCD that is not provided integral to the INVERTER, as allowed by 4.8.3.4, the installation instructions shall state the need for the RCD,.		N/A
	and shall specify its rating, type, and required circuit location		N/A
5.3.2.10	Remote indication of faults		P
	The installation instructions shall include an explanation of how to properly make connections to (where applicable), and use, the electrical or electronic fault indication required by 13.9.		P
5.3.2.11	External array insulation resistance measurement and response		N/A
	The installation instructions for an INVERTER for use with ungrounded arrays that does not incorporate all the aspects of the insulation resistance measurement and response requirements in 4.8.2.1, must include:		N/A
	- for isolated INVERTERs: an explanation of what aspects of array insulation resistance measurement and response are not provided, and		N/A
	- an instruction to consult local regulations to determine if any additional functions are required or not;		N/A
	- for non-isolated INVERTERs: an explanation of what external equipment must be provided in the system, and		N/A
	- what the setpoints and response implemented by that equipment must be, and:		N/A
	- how that equipment is to be interfaced with the rest of the system.		N/A
5.3.2.12	Array functional grounding information		N/A
	Where approach a) of 4.8.2.2 is used, the installation instructions for the INVERTER shall include all of the following:		N/A
	a) the value of the total resistance between the PV circuit and ground integral to the INVERTER ...		N/A
	b) the minimum array insulation resistance to ground that system designer or installer must meet when selecting the PV panel and system design, based on the minimum value that the design of the PV functional grounding in the INVERTER was based on		N/A
	c) the minimum value of the total resistance $R = V_{MAX} PV/30 \text{ mA}$ that the system must meet, with an explanation of how to calculate the		N/A
	total		

	d) a warning that there is a risk of shock hazard if the total minimum resistance requirement is not met.		N/A
5.3.2.13	Stand-alone INVERTERs for dedicated loads		N/A
	Where the approach of 4.7.5.5 is used, the installation instructions for the INVERTER shall include a warning that the INVERTER is only to be used with the dedicated load for which it was evaluated, and		N/A
	shall specify the dedicated load.		N/A
5.3.2.14	Identification of firmware version(s)		P
	An INVERTER utilizing firmware for any protective functions shall provide means to identify the firmware version.		P
	This can be a marking, but the information can also be provided by a display panel, communications port or any other type of user interface.....	Provided by display panel when the PCE started.	P
7	Protection against electric shock and energy hazards		P
7.3	Protection against electric shock		P
7.3.10	Additional requirements for stand-alone INVERTERs		N/A
	One circuit conductor bonded to earth to create a grounded conductor and an earthed system.		N/A
	The means used to bond the grounded conductor to protective earth provided within the INVERTER or		N/A
	as part of the installation		N/A
	If not provided integral to the INVERTER, the required means shall be described in the installation instructions as per 5.3.2.8.		N/A
	The means used to bond the grounded conductor to protective earth shall comply with the requirements for protective bonding in Part 1,		N/A
	If the bond can only ever carry fault currents in stand-alone mode, the maximum current for the bond is determined by the INVERTER maximum output fault current.		N/A
	Output circuit bonding arrangements shall ensure that in any mode of operation, the system only has the grounded circuit conductor bonded to earth in one place at a time..		N/A
	Switching arrangements may be used, in which case the switching device used is to be subjected to the bond impedance test along with the rest of the bonding path		N/A
	INVERTERs intended to have a circuit conductor bonded to earth shall not impose any normal current on the bond except for leakage current.		N/A

	Outputs that are intentionally floating with no circuit conductor bonded to ground, must not have any		N/A
	voltages with respect to ground that are a shock hazard in accordance with Clause 7 of Parts 1 and 2.		
	The documentation for the INVERTER shall indicate that the output is floating as per 5.3.2.8.		N/A
7.3.11	Functionally grounded arrays		N/A
	All PV conductors in a functionally grounded array shall be treated as being live parts with respect to protection against electric shock.		N/A
9	Protection against fire hazards		P
9.3	Short-circuit and overcurrent protection		P
9.3.4	INVERTER backfeed current onto the array		P
	The backfeed current testing and documentation requirements in Part 1 apply, including but not limited to the following.		P
	INVERTER backfeed current onto the PV array maximum value.....		P
	This INVERTER backfeed current value shall be provided in the installation instructions regardless of the value of the current, in accordance with Table 33.		P
13	Physical requirements		P
13.9	Fault indication		P
	Where this Part 2 requires the INVERTER to indicate a fault, both of the following shall be provided:		P
	a) a visible or audible indication, integral to the INVERTER, and detectable from outside the INVERTER, and		P
	b) an electrical or electronic indication that can be remotely accessed and used.		P
	The installation instructions shall include information regarding how to properly make connections (where applicable) and use the electrical or electronic means in b) above, in accordance with 5.3.2.10.		P

4.4.4		TABLE: Single fault condition to be applied					P
		Ambient temperature (°C)			24°C ~ 27 °C		
		Relative humidity.....:			60%RH		
4.4.4.15.1		Fault-tolerance of residual current monitoring					
No.	component	Fault	Input (Vdc)	Output (Vac)	Test duration	Fuse current (A)	Observation
Short-circuit before energized							
1	Q3 (d-s), (AC EMI board)	Short	PV Simulator used with settings Vmax = 1000 Vdc, Isc=20 Ad.c.	1PH, 220 Va.c.	10 minutes	--	Do not connect to AC mains unless remove fault condition. The red LED was light and LCD indicate fault. No damage. No hazard.
2	Q3 (g-d), (AC EMI board)	Short		1PH, 220 Va.c.	10 minutes	--	Same as above;
3	Q3 (g-s), (AC EMI board)	Short		1PH, 220 Va.c.	10 minutes	--	Same as above;
4	Q3 pin d, (AC EMI board)	Open		1PH, 220 Va.c.	10 minutes	--	Same as above;
5	HCT7, pin 3 and pin 4, (AC EMI board)	Short		1PH, 220 Va.c.	10 minutes	--	Same as above;
6	HCT7, pin 3 and pin 2, (AC EMI board)	Short		1PH, 220 Va.c.	10 minutes	--	Same as above;
7	HCT7, pin 3 and pin 1, (AC EMI board)	Short		1PH, 220 Va.c.	10 minutes	--	Same as above;
8	HCT7, pin 4 and pin 2, (AC EMI board)	Short		1PH, 220 Va.c.	0 minutes	--	Same as above;

9	HCT7, pin 4 and pin 1, (AC EMI board)	Short	1PH, 220 Va.c.	10 minutes	--	Same as above;
10	HCT7, pin 3, (AC EMI board)	Open	1PH, 220 Va.c.	10 minutes	--	Same as above;
11	HCT7, pin 4	Open	1PH, 220 Va.c.	10 minutes	--	Same as above;
12	U1, pin 3 and pin 5, (AC EMI board)	Short	1PH, 220 Va.c.	10 minutes	--	Operating as normal, passes testing in accordance with 4.8.3.5 showing that the residual current monitoring system functions properly under the single fault condition, and indicates a fault in accordance with 13.9,

Supplementary information:

4.4.4	TABLE: Single fault condition to be applied					P
	Ambient temperature (°C) : 21 °C ~ 23 °C					
	Power source for EUT: Manufacturer, model/type, output rating : --					
4.4.4.15.2	Fault-tolerance of automatic disconnecting means					
Component No.	Fault	Supply voltage (V)	Test time	Fuse #	Fuse current (A)	Observation
Relay K1, contact	Each Relay contact short-circuit one at a time before energized PV and AC	PV Simulator used with settings Vmax = 1000 Vdc, Isc=20 Ad.c. x 3	10 minutes	--	--	LCD indicates fault "Relay fault", Does not connect to AC mains, can reset by remove fault condition. See below picture.
Relay K2, contact			10 minutes	--	--	Same as above
Relay K3, contact			10 minutes	--	--	Same as above
Relay K4, contact			10 minutes	--	--	Same as above
Relay K5, contact			10 minutes	--	--	Same as above
Relay K6, contact			10 minutes	--	--	Same as above

Check that the relays fulfil the basic insulation or simple separation based on the PV circuit working voltage.	The remaining relay gaps in series provided basic clearance: 2,0 mm
Each active phase can be switched. (L and N)	Three Line conductors disconnected simultaneously.
Supplementary information: Analysis of the design indicates that all the models can be expected to have the same result.	

4.4.4.17	Cooling system failure – Blanketing test	P
	Test voltage (Vdc)	Vmax = 1000 Vdc
	Test current (Idc)	Isc=20 Ad.c. x 3
	Test voltage (Vac)	1PH, 220Va.c.
	Test current (Iac)	Output full load
	tamb1 (°C)	24,0 °C
	tamb2 (°C)	27,5 °C
	Test duration.....	7 hours
maximum temperature T of part/at::		T (°C)
Outer surface, top		86,3
Outer surface, side		78,8
Outer surface, front		76,9
Supplementary information: N/A		

4.8.2	TABLE: Array insulation resistance detection for INVERTERs for ungrounded and functionally grounded arrays				P
4.8.2.1	Array insulation resistance detection for INVERTERs for ungrounded arrays				P
DC Voltage below minimum operating voltage (V)	DC Voltage for INVERTER begin operation (V)	Resistance between ground and PV input terminal (Ω)	Required Insulation resistance $R = (V_{MAX} PV / 30mA) (\Omega)$	Result	
DC+ (PV Simulator used with settings $V_{max} = 1000 V_{dc}$, $I_{sc}=20 Ad.c. \times 3$)					
120Vdc	180 Vdc	>1 M Ω	1000 Vdc / 30 mA=33,33 k Ω	LCD indicates fault. The unit does not connect to AC mains, can reset by remove fault condition. See below picture.	
DC- (PV Simulator used with settings $V_{max} = 1000 V_{dc}$, $I_{sc}=20 Ad.c. \times 3$)					

120 Vdc	180 Vdc	>1 MΩ	1000 Vdc / 30 mA=33,33 kΩ	As above
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Note:

For isolated INVERTERs, shall indicate a fault in accordance with 13.9 (operation is allowed); the fault indication shall be maintained until the array insulation resistance has recovered to a value higher than the limit above. For non-isolated INVERTERs, or INVERTERs with isolation not complying with the leakage current limits in the minimum INVERTER isolation requirements in Table 30, shall indicate a fault in accordance with 13.9, and shall not connect to the mains; the INVERTER may continue to make the measurement, may stop indicating a fault and may connect to the mains if the array insulation resistance has recovered to a value higher than the limit above. It is not required to test all PV input terminals if analysis of the design indicates that one or more terminals can be expected to have the same result, for example where multiple PV string inputs are in parallel.

Supplementary information:

	Tracker 1 DC+	Tracker 1 DC-	Tracker 2 DC+	Tracker 2 DC-	Tracker 3 DC+	Tracker 3 DC-
Trip settings (kΩ)	40000 kΩ	40000 kΩ	40000 kΩ	40000 kΩ	40000 kΩ	40000 kΩ
Measured trigger point (kΩ)	40023 kΩ	40035 kΩ	40026 kΩ	40028 kΩ	40024 kΩ	40031 kΩ
Measured recover point (kΩ)	40028 kΩ	40039 kΩ	40030 kΩ	40031 kΩ	40035 kΩ	40038 kΩ

4.8.3.5	TABLE: Protection by residual current monitoring								P
Test conditions:		See below							
4.8.3.5.2	Test for detection of excessive continuous residual current								P
Tracker No.	PV (+ or -)	Input (Vdc)	Output (Vac, kW)	Baseline trip current (mA)	Measured trigger time (ms) shall < 300 ms (repeat 5 times)				
1	+	450 V	1PH, 220 Va.c., 5.5 kW	230 mA	256 ms	272 ms	280 ms	218 ms	232 ms
1	-	450 V	1PH, 220 Va.c., 5.5 kW	211 mA	216 ms	204 ms	220 ms	270 ms	221 ms

Note:

maximum 300mA for INVERTERs with continuous output power rating ≤30 kVA;
maximum 10mA per kVA of rated continuous output power for INVERTERs with continuous output power rating > 30 kVA.

This test shall be repeated 5 times, and for all 5 tests the time to disconnect shall not exceed 0,3s.

The test is repeated for each PV input terminal. It is not required to test all PV input terminals if analysis of the design indicates that one or more terminals can be expected to have the same result, for example where multiple PV string inputs are in parallel.

4.8.3.5	TABLE: Protection by residual current monitoring								P
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Test conditions:		See below							
4.8.3.5.2		Test for detection of excessive continuous residual current							P
Tracker No.	PV (+ or -)	Input (Vdc)	Output (Vac, kW)	Baseline trip current (mA)	Measured trigger time (ms) shall < 300 ms (repeat 5 times)				
1	+	450 V	1PH, 220 Va.c., 5.5 kW	230 mA	256 ms	272 ms	280 ms	218 ms	232 ms
1	-	450 V	1PH, 220 Va.c., 5.5 kW	211 mA	216 ms	204 ms	220 ms	270 ms	221 ms
Note: maximum 300mA for INVERTERs with continuous output power rating ≤ 30 kVA; maximum 10mA per kVA of rated continuous output power for INVERTERs with continuous output power rating > 30 kVA. The capacitive current is risen until disconnection. Test condition: $I_c + 30/60/150\text{mA} \leq I_{\text{cmax}}$. R1 is set that 30/60/150mA Flow and switch S is closed. This test shall be repeated 5 times, and for all 5 tests the time to disconnect shall not exceed 0.3s. The test is repeated for each PV input terminal. It is not required to test all PV input terminals if analysis of the design indicates that one or more terminals can be expected to have the same result, for example where multiple PV string inputs are in parallel.									

4.8.3.5		TABLE: Protection by residual current monitoring							P		
Test conditions:			See below								
4.8.3.5.3		TABLE: Test for detection of sudden changes in residual current							P		
Tracker No.	PV (+ or -)	Input (Vdc)	Output (Vac, kW)	Baseline trip current (mA)	Measured trigger time (ms) shall < 300 ms for 30 mA; (repeat 5 times)						
30 mA sudden changes in residual current											
1	+	450 V	1PH, 220 Va.c., 5.5 kW	230 mA	202 ms	218 ms	216 ms	234 ms	218 ms		
1	-	450 V	1PH, 220 Va.c., 5.5 kW	211 mA	240 ms	256 ms	200 ms	214 ms	217 ms		
Note: The capacitive current is risen until disconnection. Test condition: $I_c + 30/60/150\text{mA} \leq I_{\text{cmax}}$. R1 is set that 30/60/150mA Flow and switch S is closed. Supplementary information: The unit have three independent MPP trackers, analysis of the design indicates that any one PV + or PV – terminal can be expected to have the same result. All above tests indicate a fault in accordance with clause 13.9 as below LCD display;											

4.8.3.5	TABLE: Protection by residual current monitoring								P
Test conditions:		See below							
4.8.3.5.3	TABLE: Test for detection of sudden changes in residual current								P
Tracker No.	PV (+ or -)	Input (Vdc)	Output (Vac, kW)	Baseline trip current (mA))	Measured trigger time (ms) shall < 150 ms for 60 mA; (repeat 5 times)				
60 mA sudden changes in residual current									
1	+	450 V	1PH, 220 Va.c., 5.5 kW	230 mA	108 ms	137 ms	142 ms	138 ms	135 ms
1	-	450 V	1PH, 220 Va.c., 5.5 kW	211 mA	130 ms	100 ms	138 ms	130 ms	134 ms

Supplementary information:

- (1) The unit have three independent MPP trackers, analysis of the design indicates that any one PV + or PV – terminal can be expected to have the same result.
- (2) All above tests indicate a fault in accordance with clause 13.9 as below LCD display;

4.8.3.5	TABLE: Protection by residual current monitoring								P
Test conditions:			See below						
4.8.3.5.3	TABLE: Test for detection of sudden changes in residual current								P
Tracker No.	PV (+ or -)	Input (Vdc)	Output (Vac, kW)	Baseline trip current (mA)	Measured trigger time (ms) shall < 40 ms for 150 mA; (repeat 5 times)				
150 mA sudden changes in residual current									
1	+	450 V	1PH, 220 Va.c., 5.5 kW	230 mA	28 ms	29 ms	32 ms	32 ms	38 ms
1	-	450 V	1PH, 220 Va.c., 5.5 kW	211 mA	18 ms	18 ms	21 ms	23 ms	29 ms

Note:

The capacitive current is risen until disconnection.

Test condition: $I_c + 30/60/150\text{mA} \leq I_{\text{cmax}}$. R1 is set that 30/60/150mA Flow and switch S is closed.

Supplementary information:

The unit have three independent MPP trackers, analysis of the design indicates that any one PV + or PV – terminal can be expected to have the same result.

All above tests indicate a fault in accordance with clause 13.9 as below LCD display;

Annex
Photo of product

